

Future Media Internet

Guoping Tan

Email: gptan@hhu.edu.cn

2023. 11. 08

Lecture 2

OSI Model & TCP/IP

Outline

- **2-1 THE OSI MODEL**
- **2-2 TCP/IP PROTOCOL SUITE**
- **2-3 5G NR Protocol (Backup)**

2-1 THE OSI MODEL

*Established in 1947, the International Standards Organization (**ISO**) is a multinational body dedicated to worldwide agreement on international standards. An ISO standard that covers all aspects of network communications is the Open Systems Interconnection (**OSI**) model. It was first introduced in the late 1970s.*

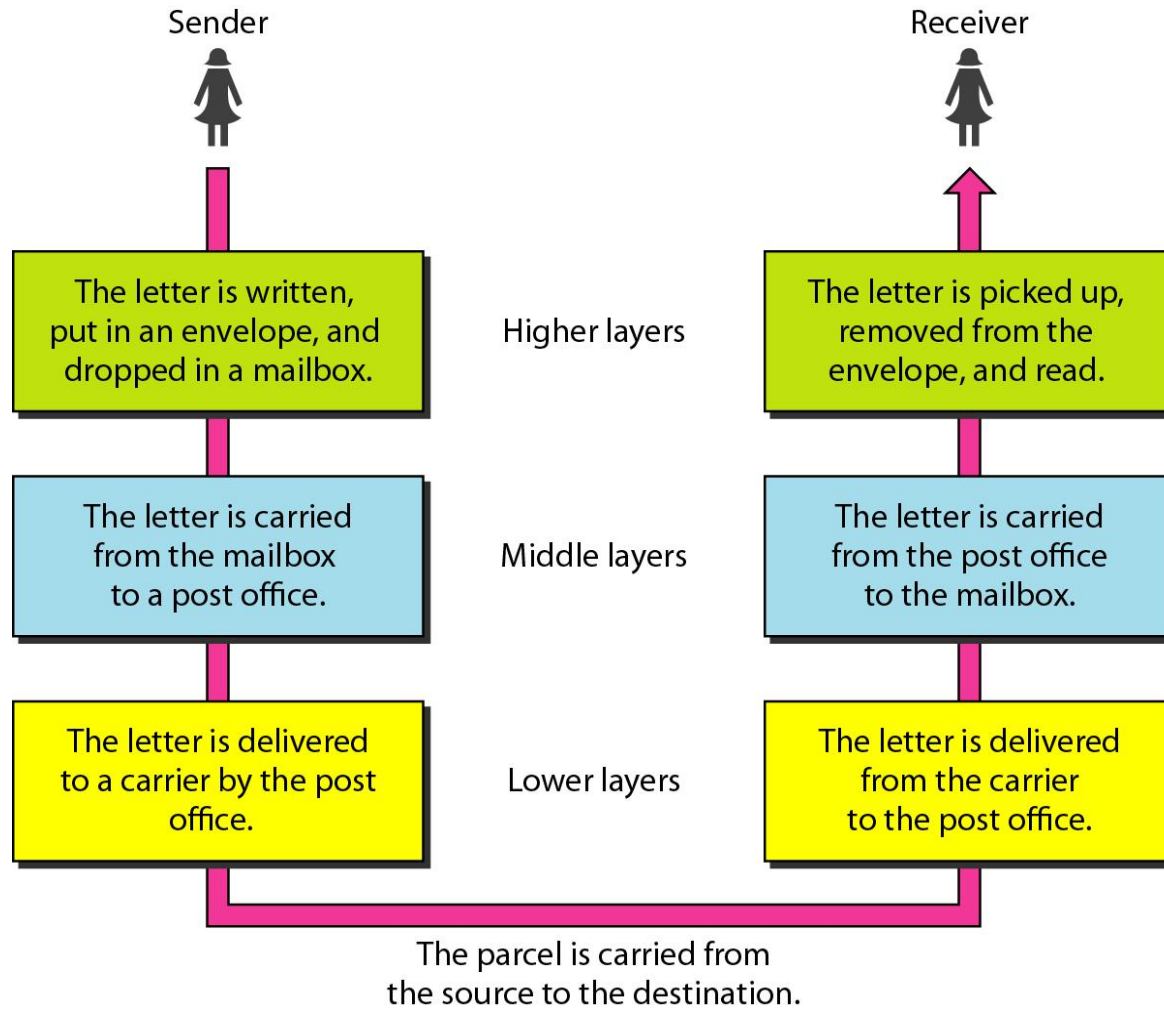
Topics discussed in this section:

Layered Architecture

Peer-to-Peer Processes

Encapsulation

Layered Tasks: Tasks involved in sending a letter

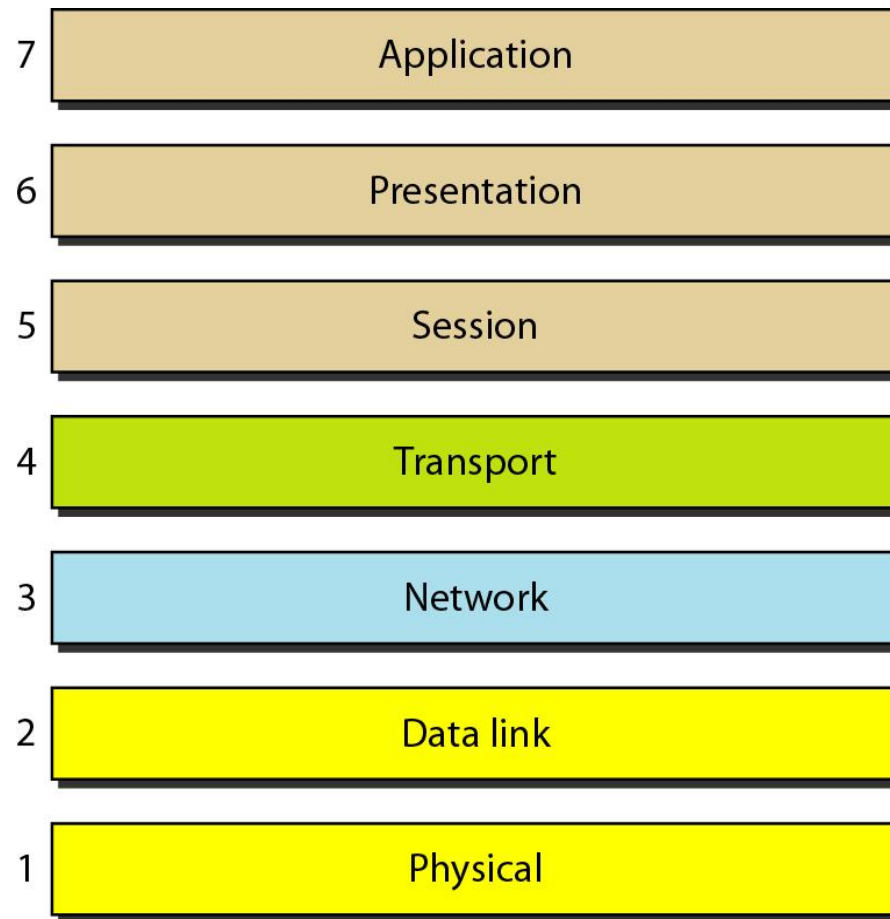




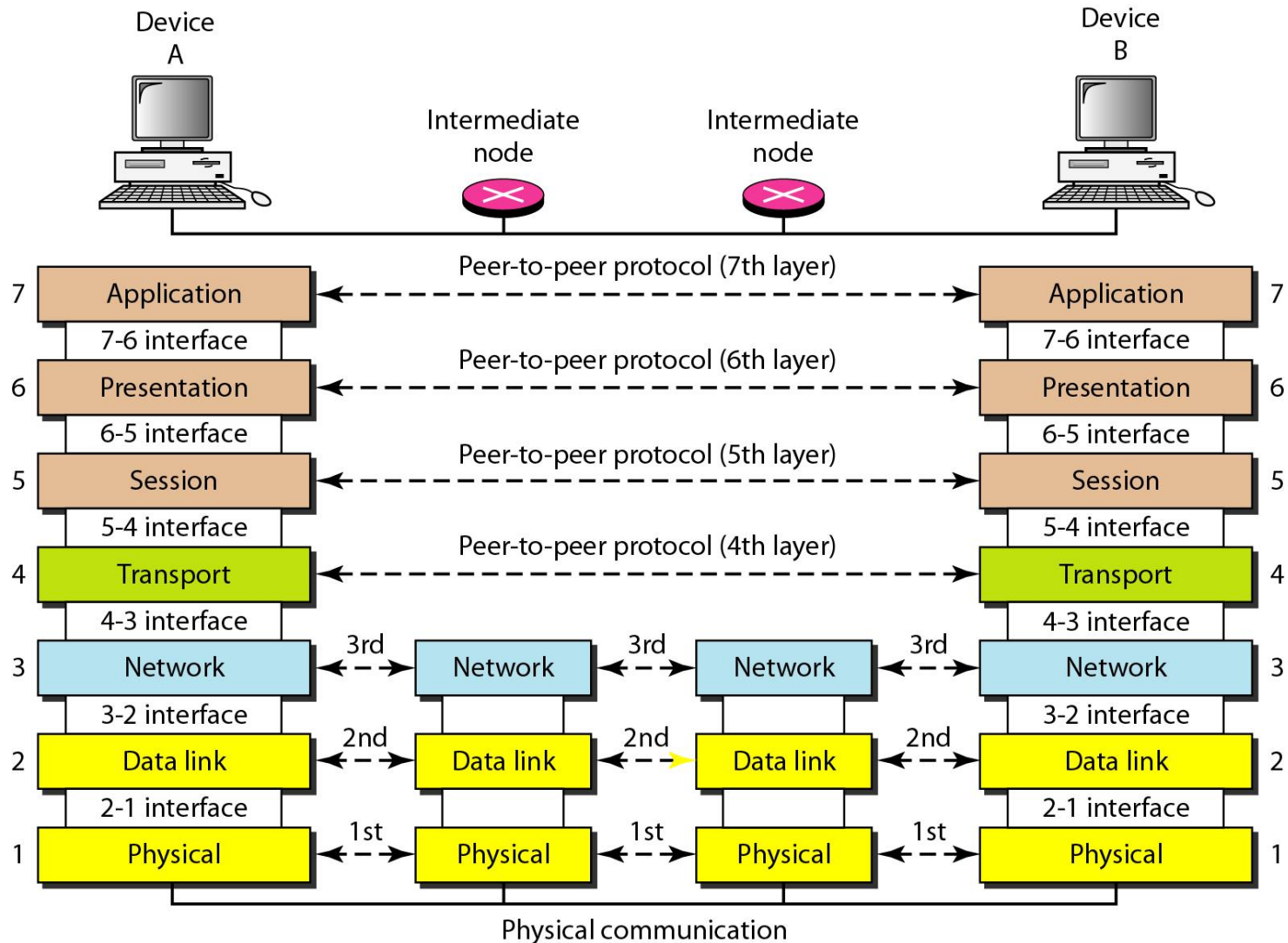
Note

**ISO is the organization.
OSI is the model.**

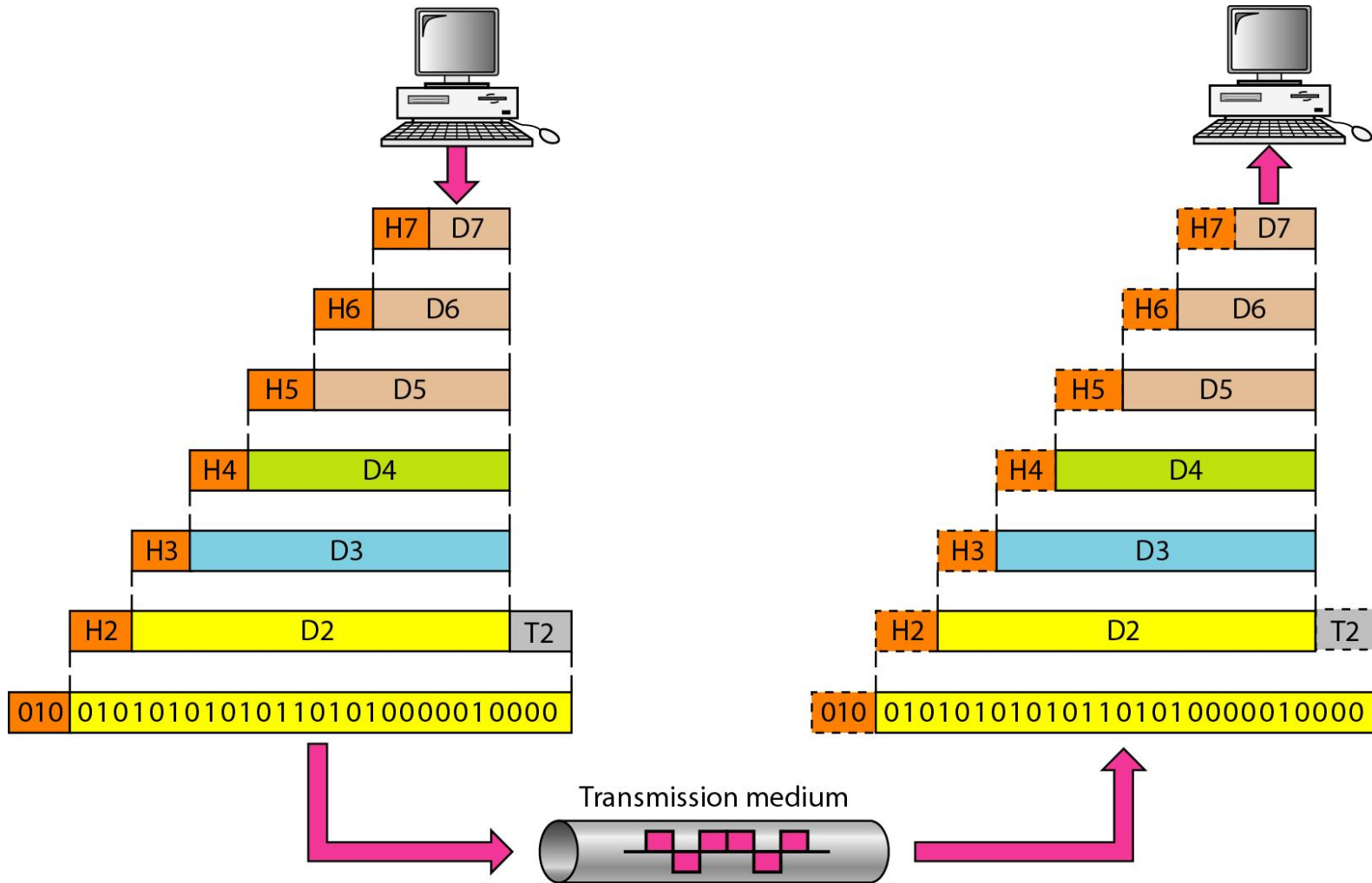
Seven layers of the OSI model



The interaction between layers in the OSI model



An exchange using the OSI model



LAYERS IN THE OSI MODEL

In this section we briefly describe the functions of each layer in the OSI model.

Topics discussed in this section:

Physical Layer

Data Link Layer

Network Layer

Transport Layer

Session Layer

Presentation Layer

Application Layer

Physical layer

The physical layer is concerned with the following:

Physical characteristics of interfaces and medium: the types of transmission medium

Representation of bits: defines the types of encoding (how 0s and 1s are changed to signal)

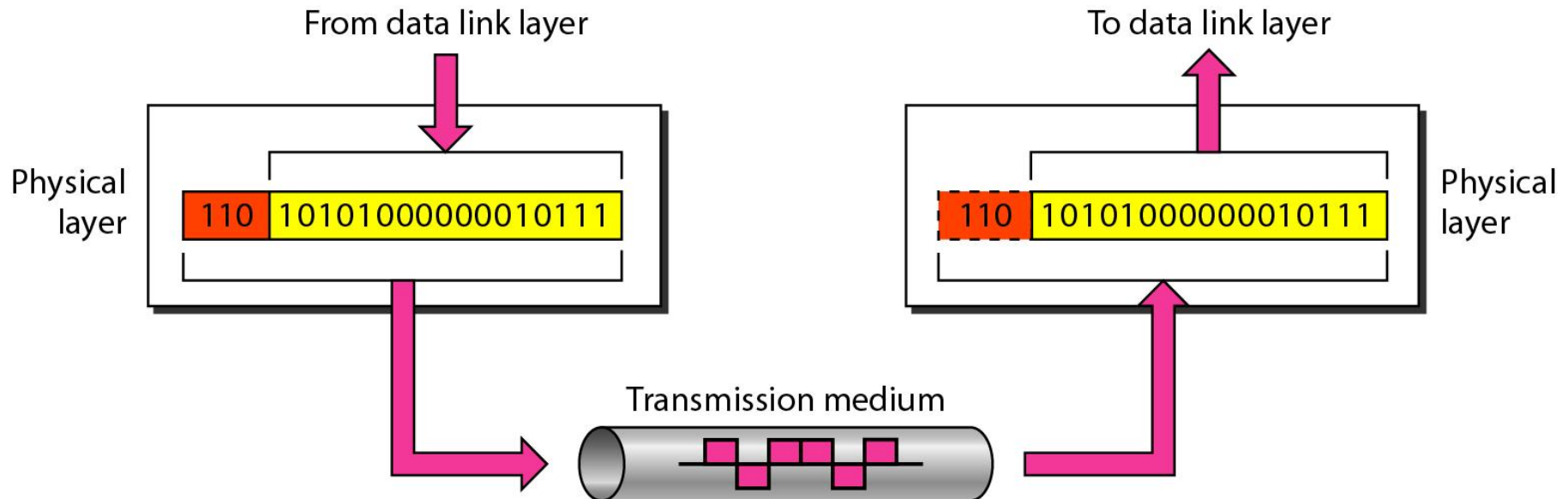
Data rate: The number of bits sent each second is also defined

Synchronization of bits: The sender and the receiver clocks must be synchronized

Line configuration: Point-to-point, Multipoint

Physical topology: Mesh, Star, Bus, Ring,...

Transmission mode: Simplex, half-duplex, full-duplex,...





Note

The physical layer is responsible for movements of individual bits from one hop (node) to the next.

Data link layer

The data link layer is concerned with the following:

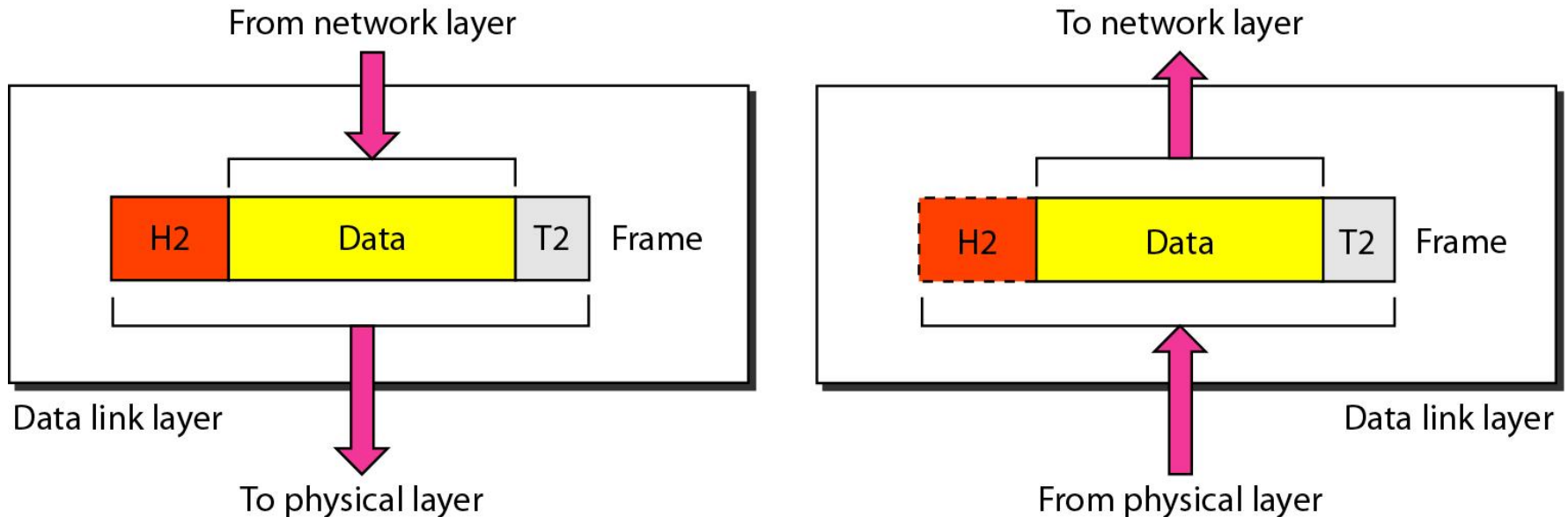
Framing: divide the stream of bits into manageable data units

Physical addressing:

Flow control: single link

Error control: single link

Access control

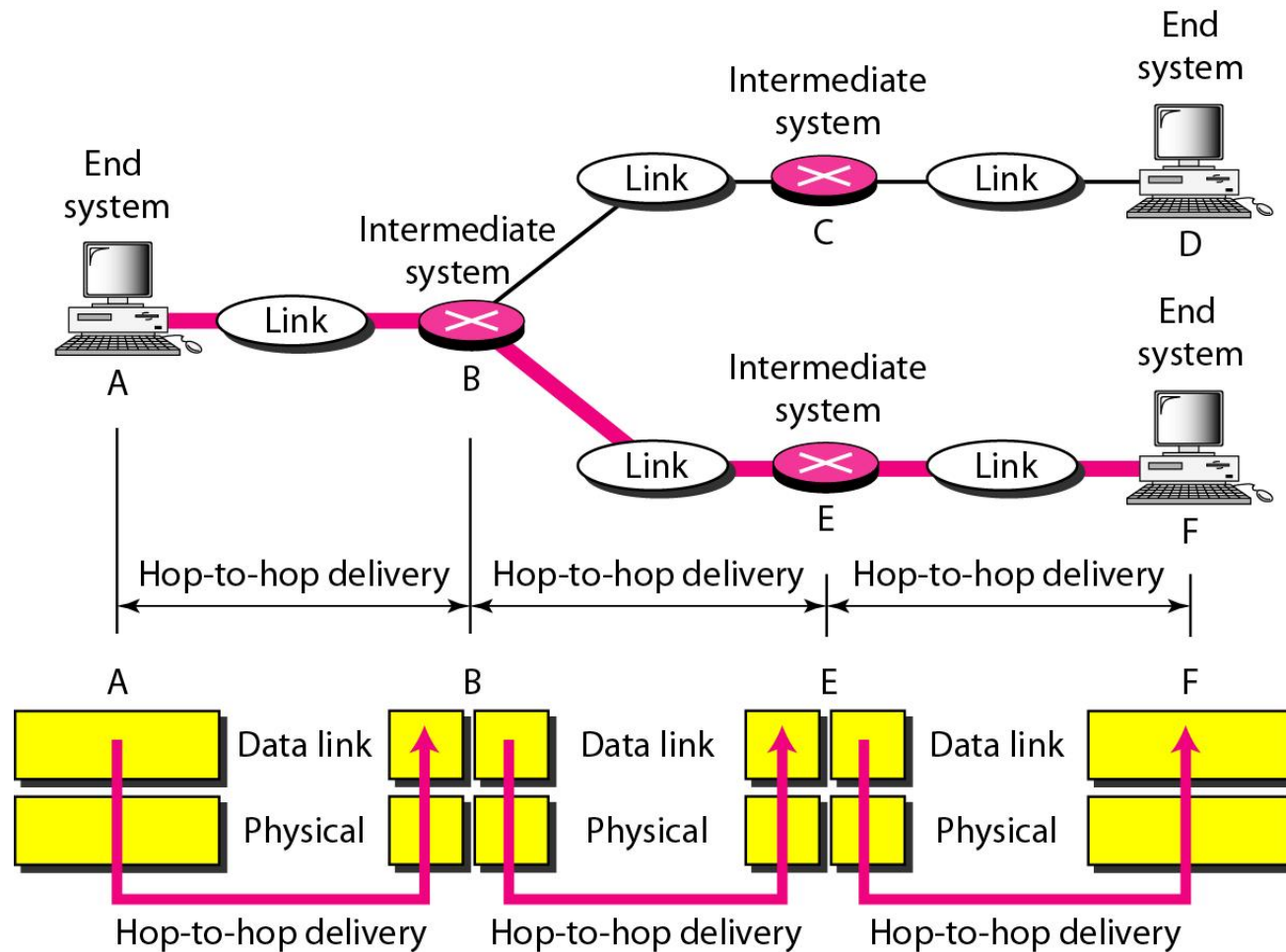




Note

The data link layer is responsible for moving frames from one hop (node) to the next.

Hop-to-hop delivery

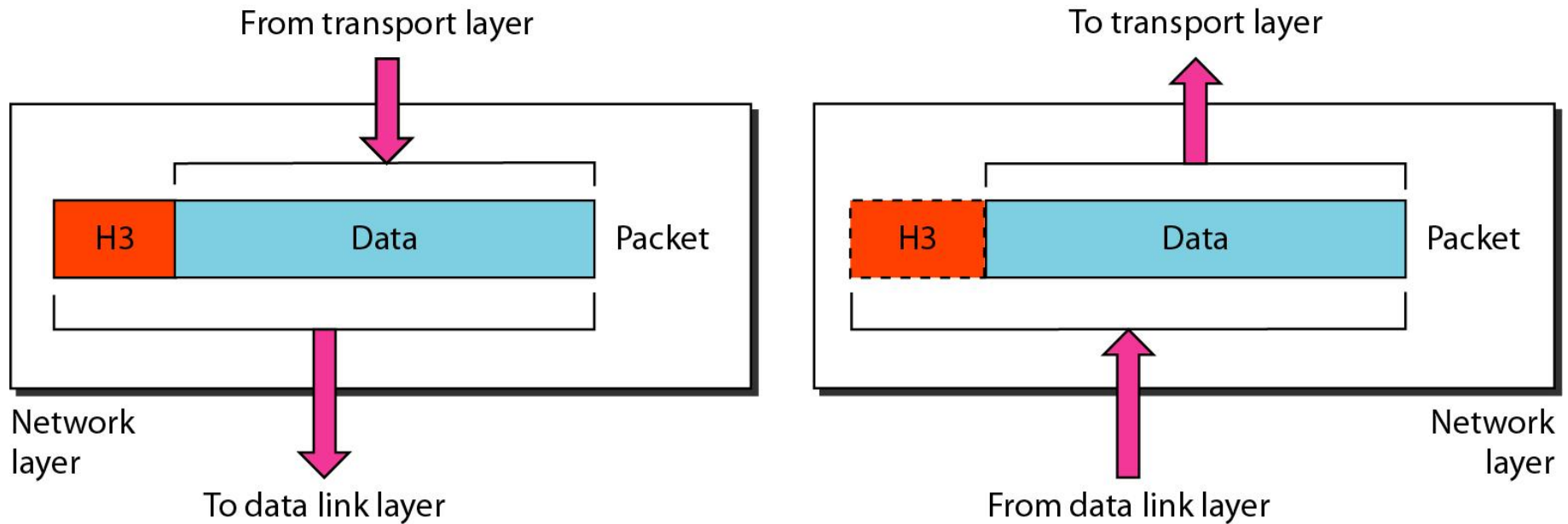


Network layer

The network layer is concerned with the following:

Logical addressing

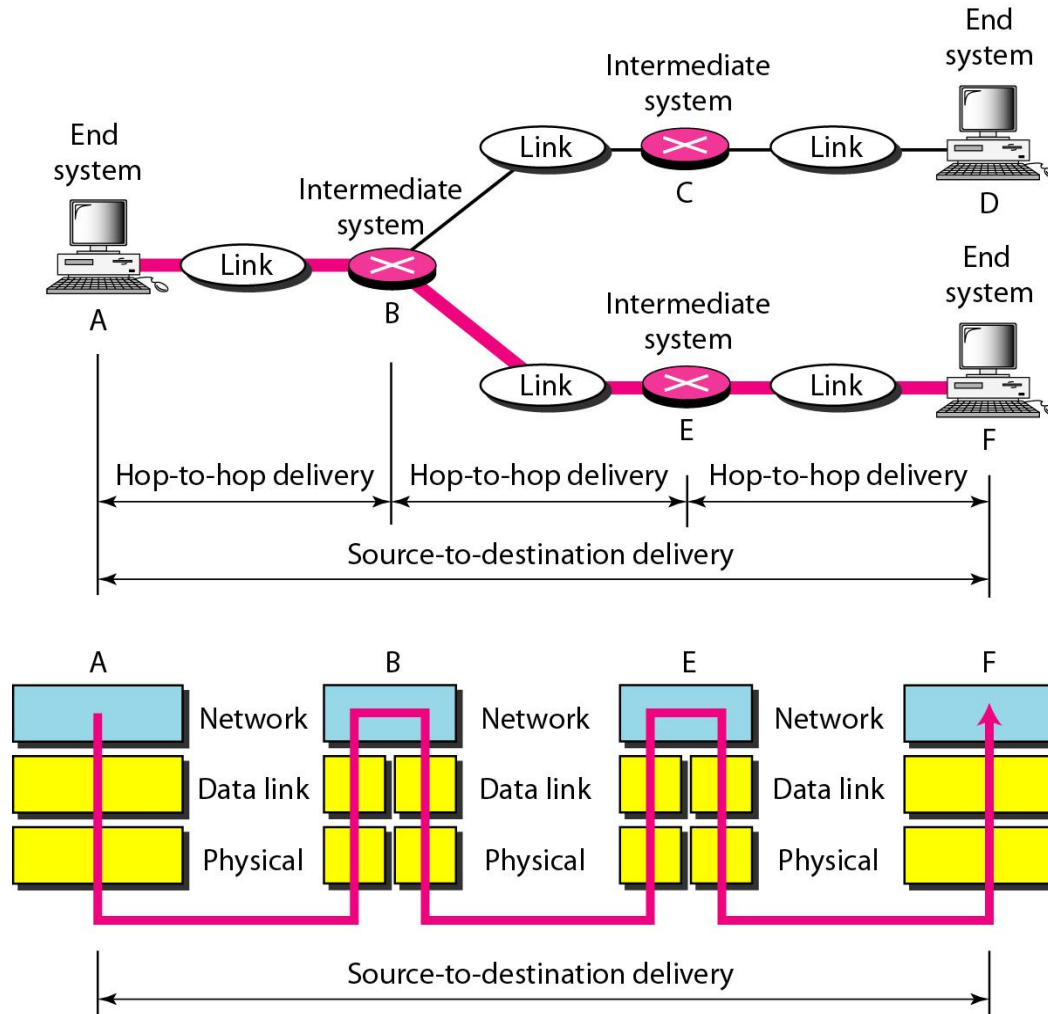
Routing



Note

The network layer is responsible for the delivery of individual packets from the source host to the destination host.

Source-to-destination delivery



Transport layer

The transport layer is concerned with the following:

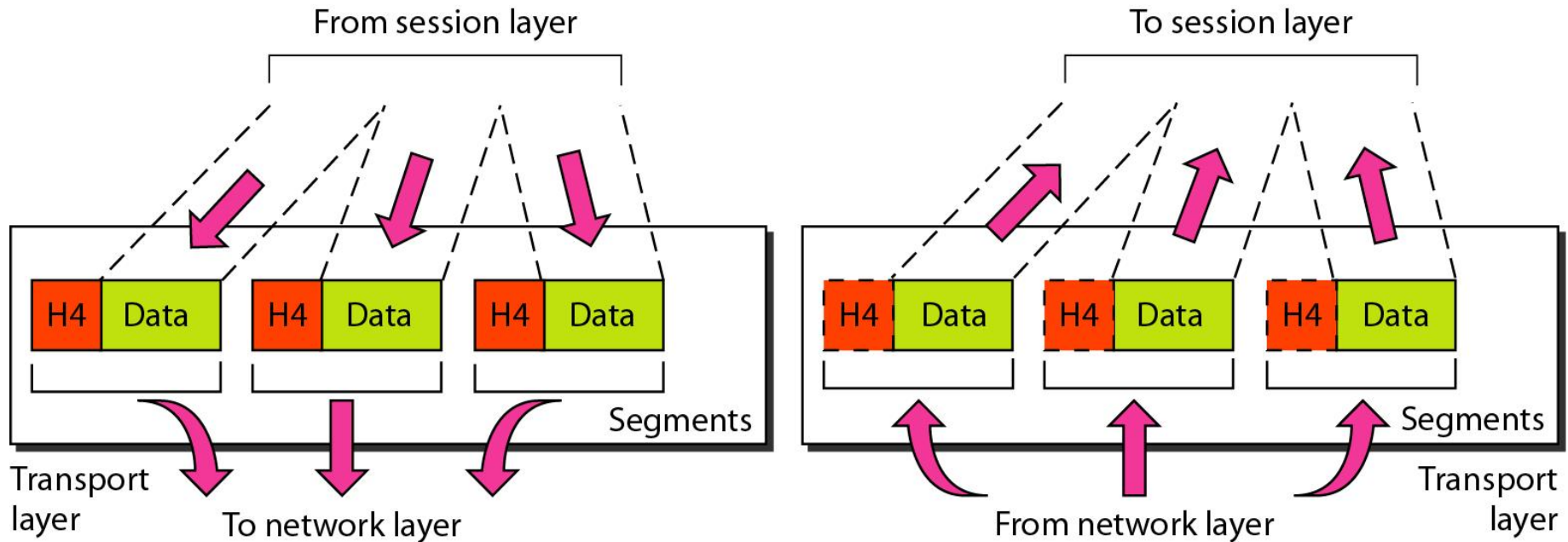
Service-point addressing

Segmentation and reassembly

Connection control

Flow control: end-to-end

Error control: process-process, rather than across a single link

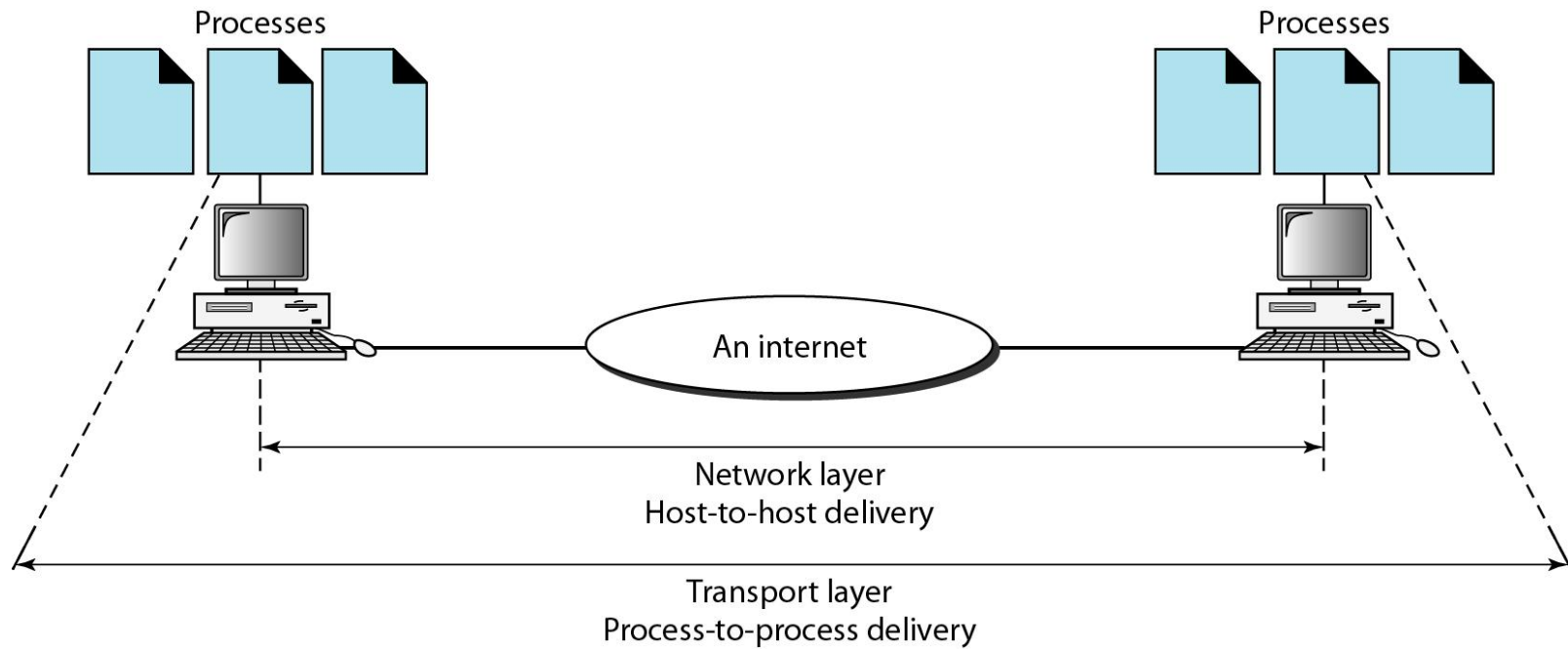




Note

The transport layer is responsible for the delivery of a message from one process to another.

Reliable process-to-process delivery of a message

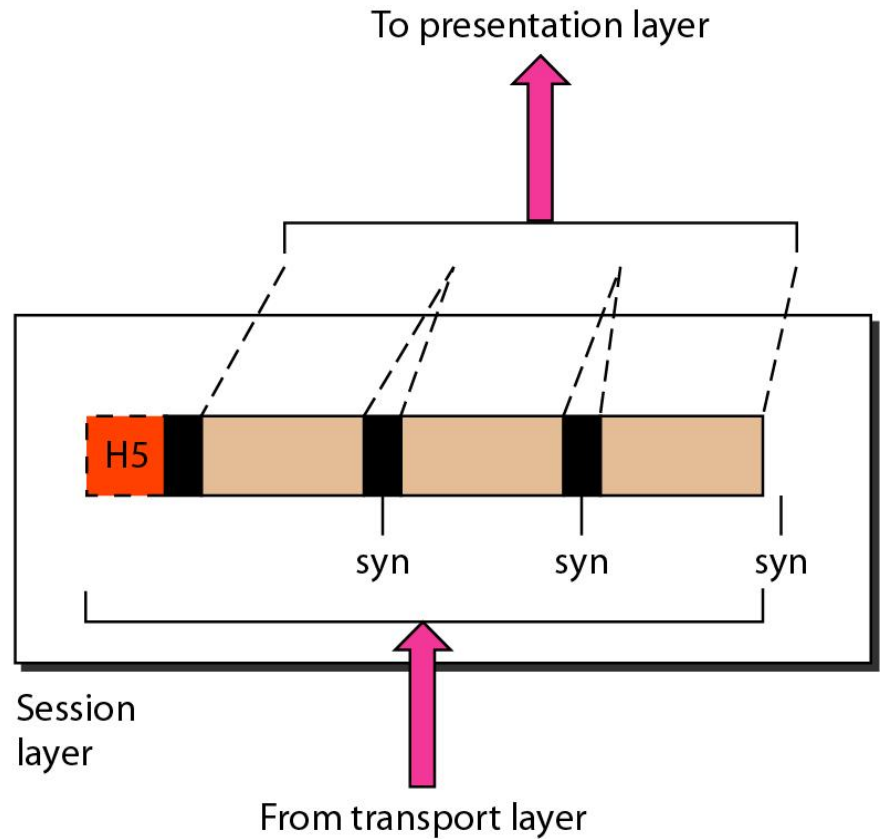
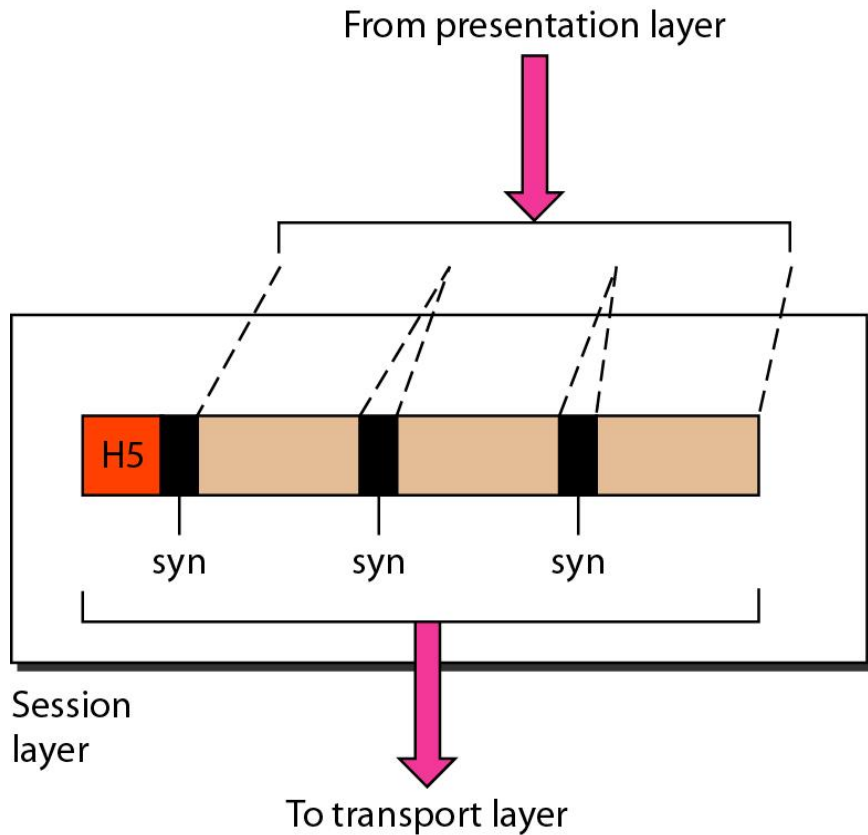


Session layer

The session layer is concerned with the following:

Dialog control

Synchronization



Note

The session layer is responsible for dialog control and synchronization.

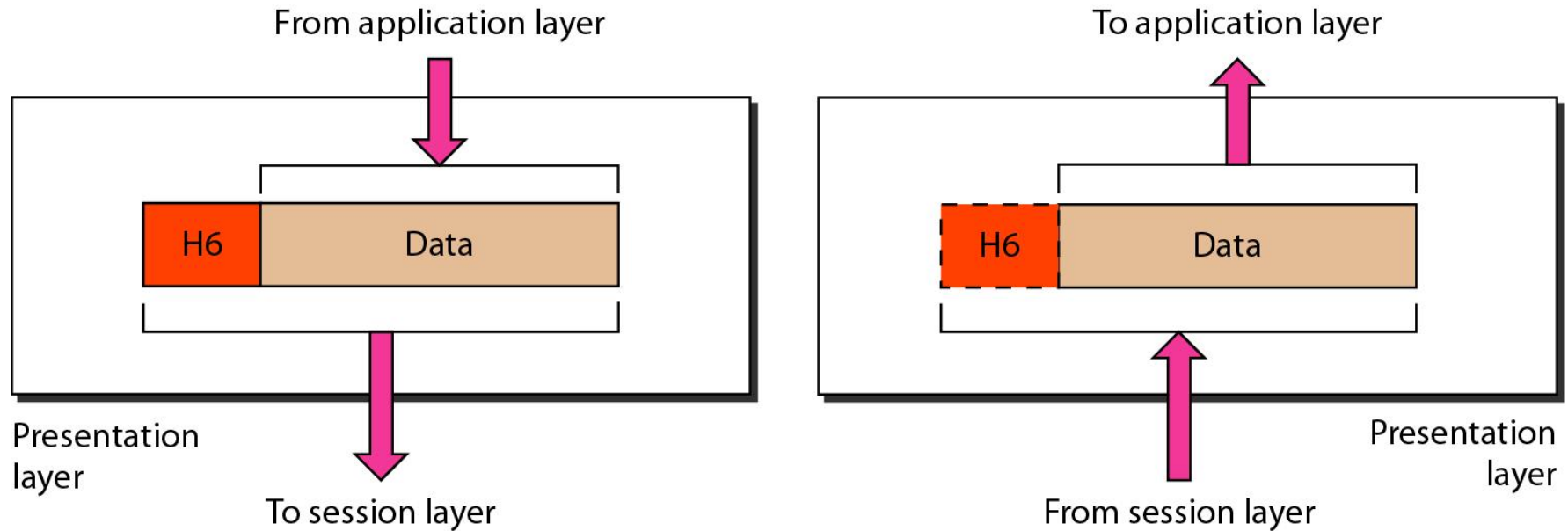
Presentation layer

The presentation layer is concerned with the following:

Translation

Encryption

Compression



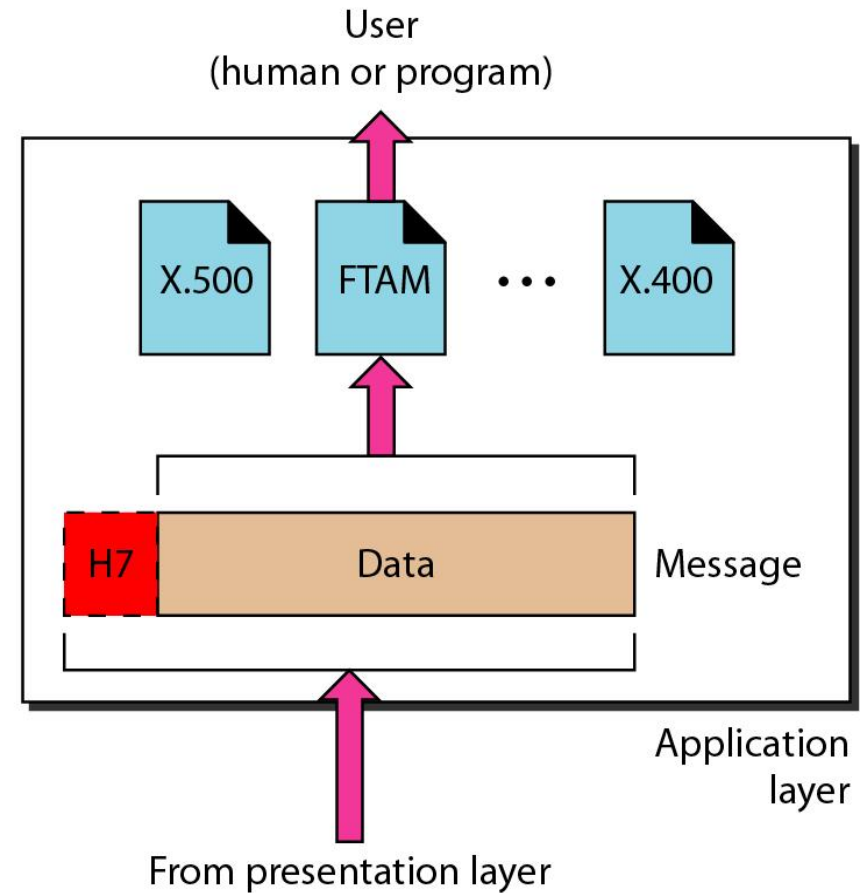
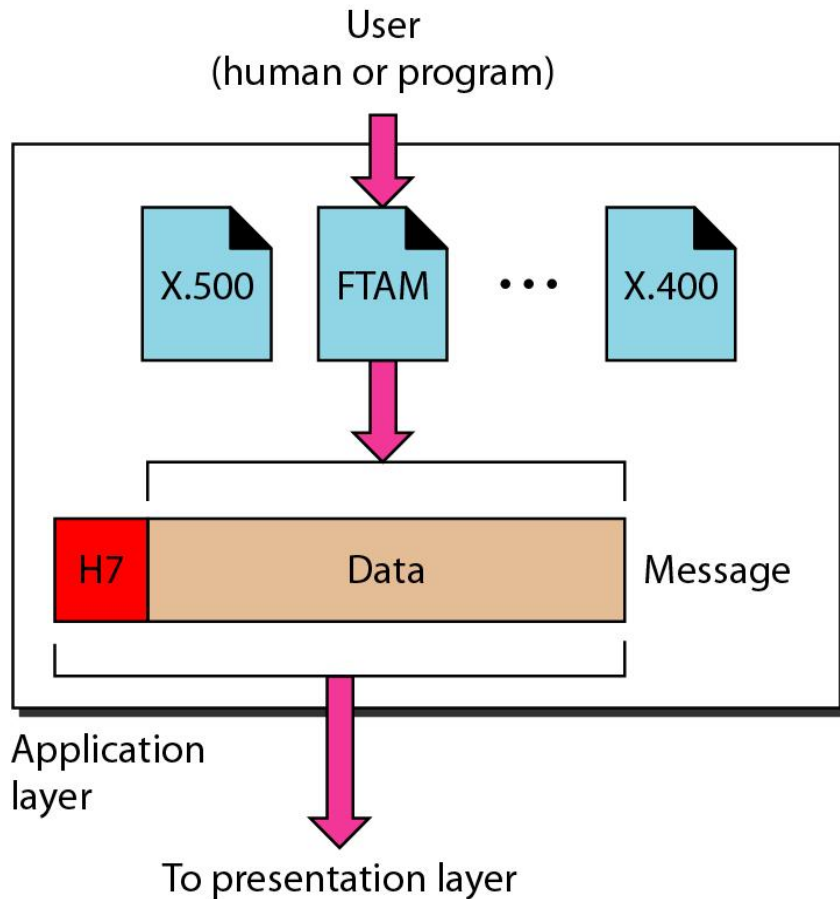


Note

The presentation layer is responsible for translation, compression, and encryption.

Application layer

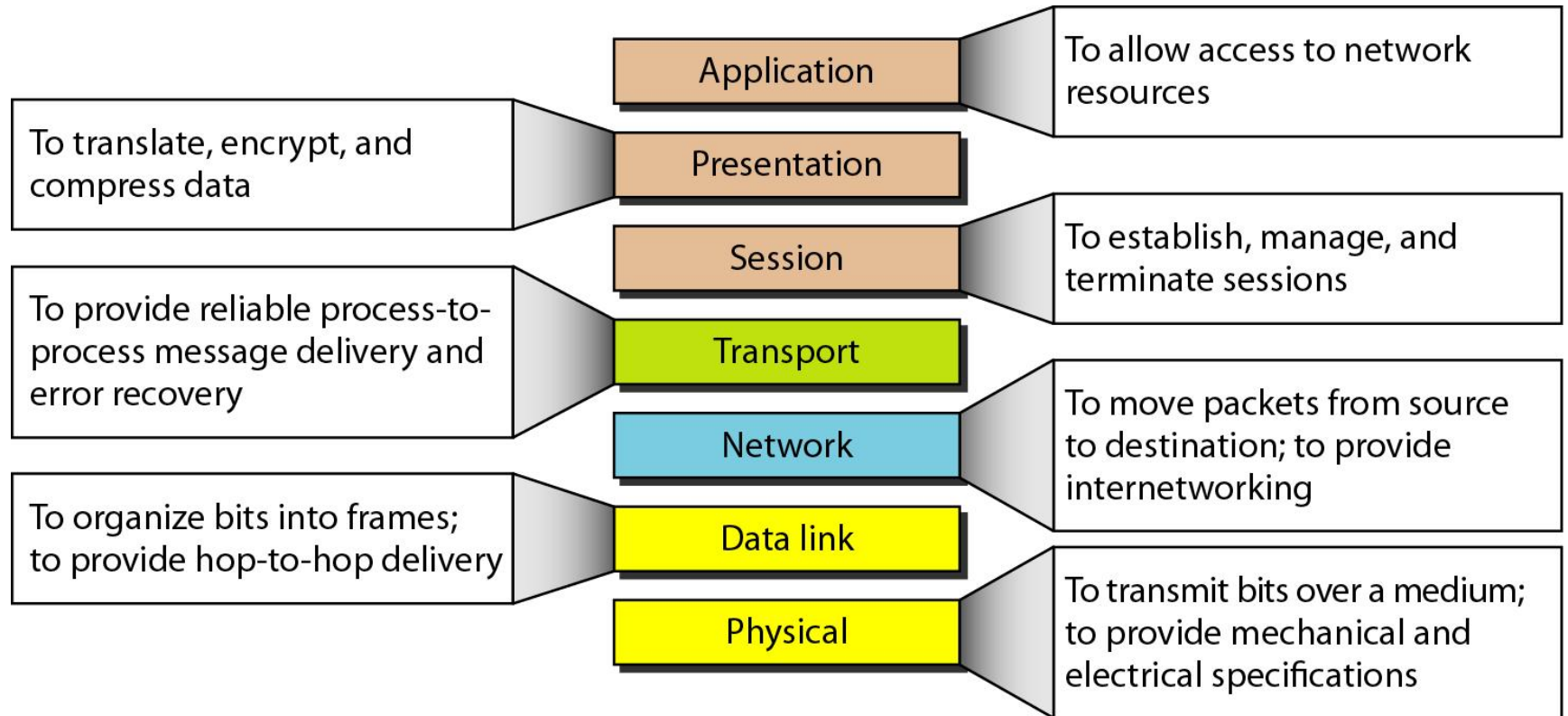
The application layer is the Network Virtual Terminal.



Note

The application layer is responsible for providing services to the user, such as File transfer, access, and management, Mail services and Directory services etc.

Summary of layers



2-2 TCP/IP PROTOCOL SUITE

*The layers in the **TCP/IP protocol suite** do not exactly match those in the OSI model. The original TCP/IP protocol suite was defined as having four layers: **host-to-network**, **internet**, **transport**, and **application**. However, when TCP/IP is compared to OSI, we can say that the TCP/IP protocol suite is made of five layers: **physical**, **data link**, **network**, **transport**, and **application**.*

Topics discussed in this section:

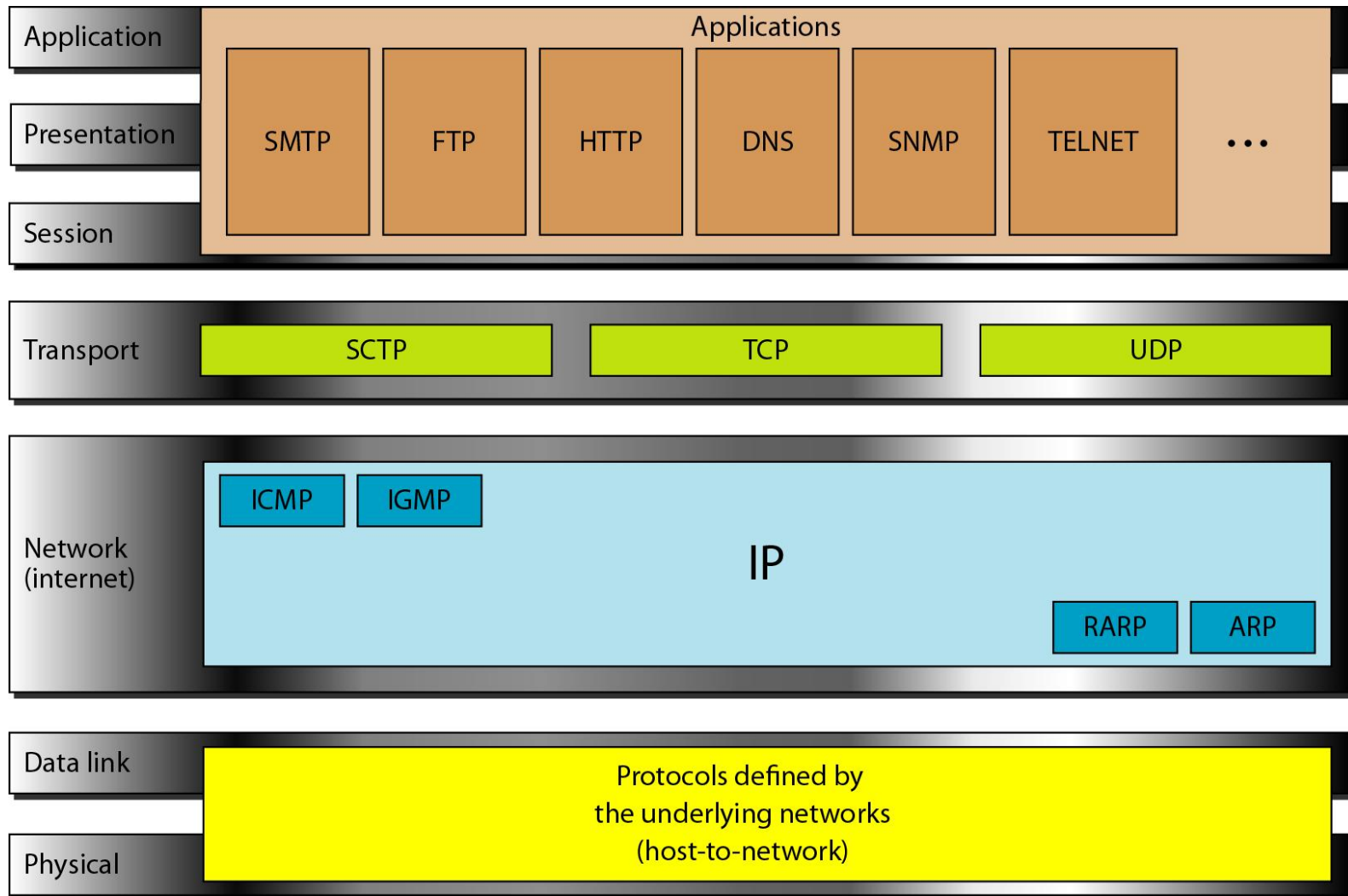
Physical and Data Link Layers

Network Layer

Transport Layer

Application Layer

TCP/IP and OSI model



ADDRESSING

*Four levels of addresses are used in an internet employing the TCP/IP protocols: **physical**, **logical**, **port**, and **specific**.*

Topics discussed in this section:

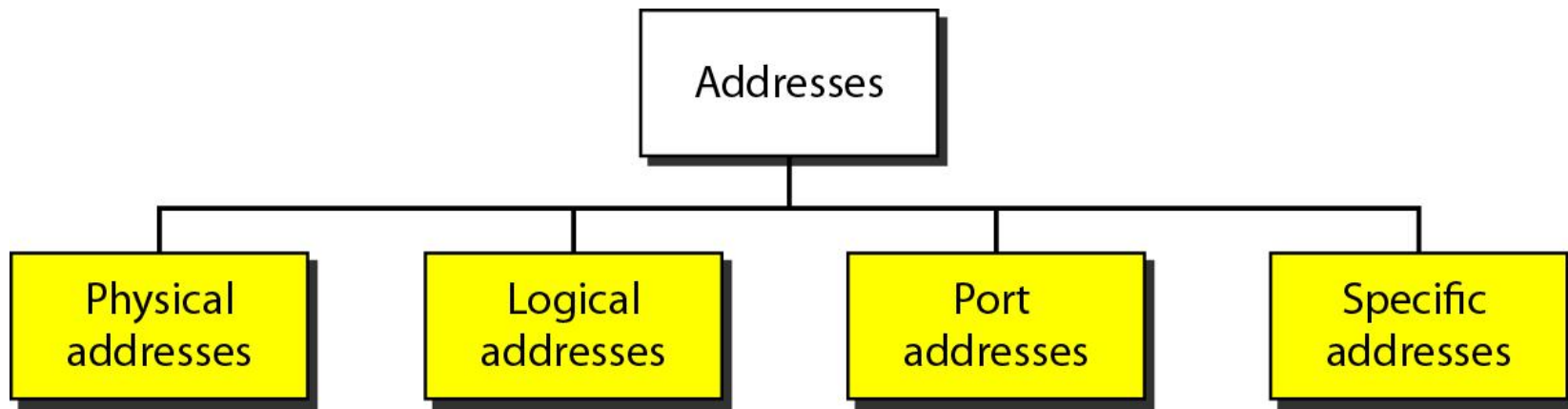
Physical Addresses

Logical Addresses

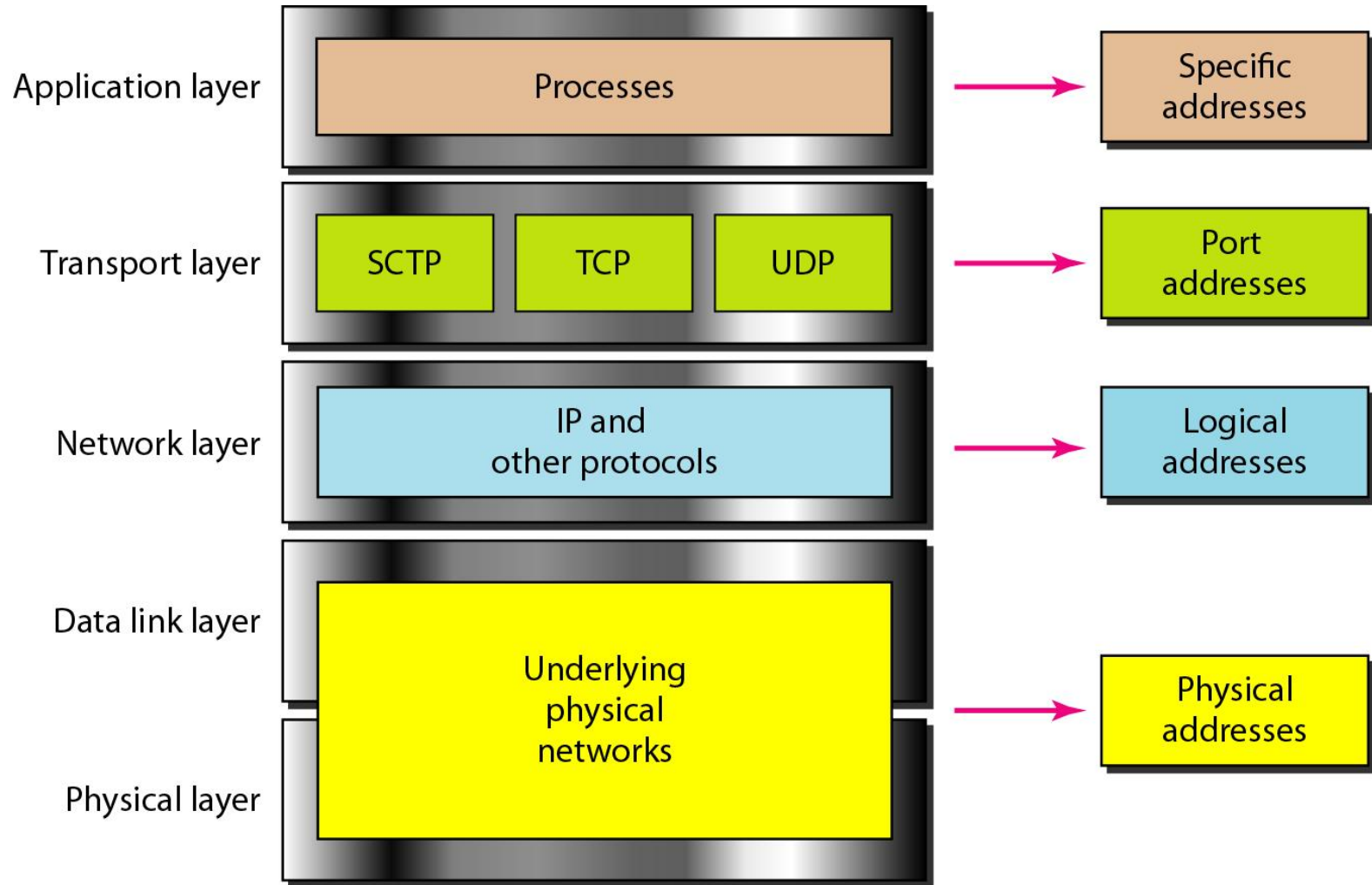
Port Addresses

Specific Addresses

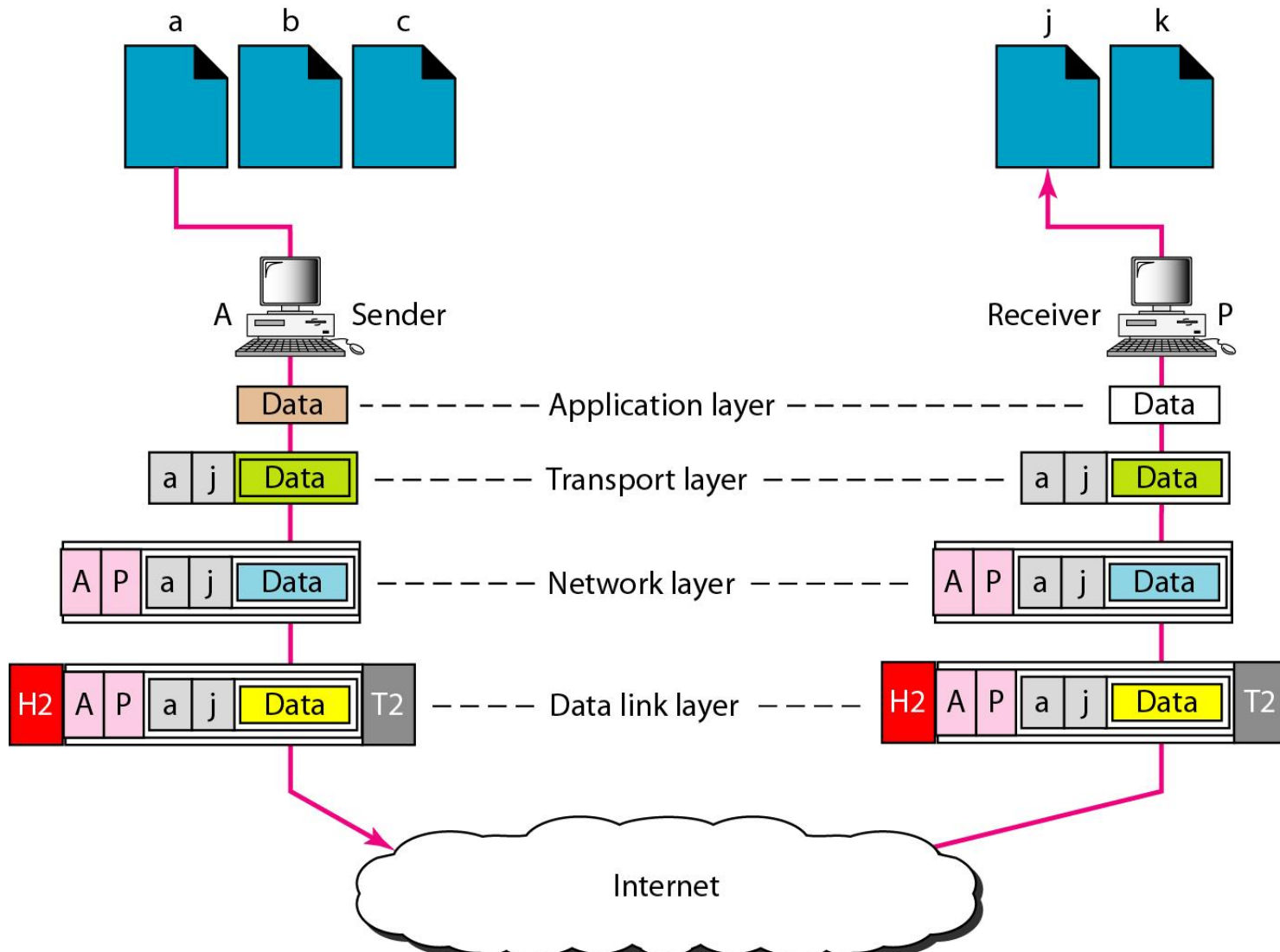
Addresses in TCP/IP



Relationship of layers and addresses in TCP/IP



Port addresses

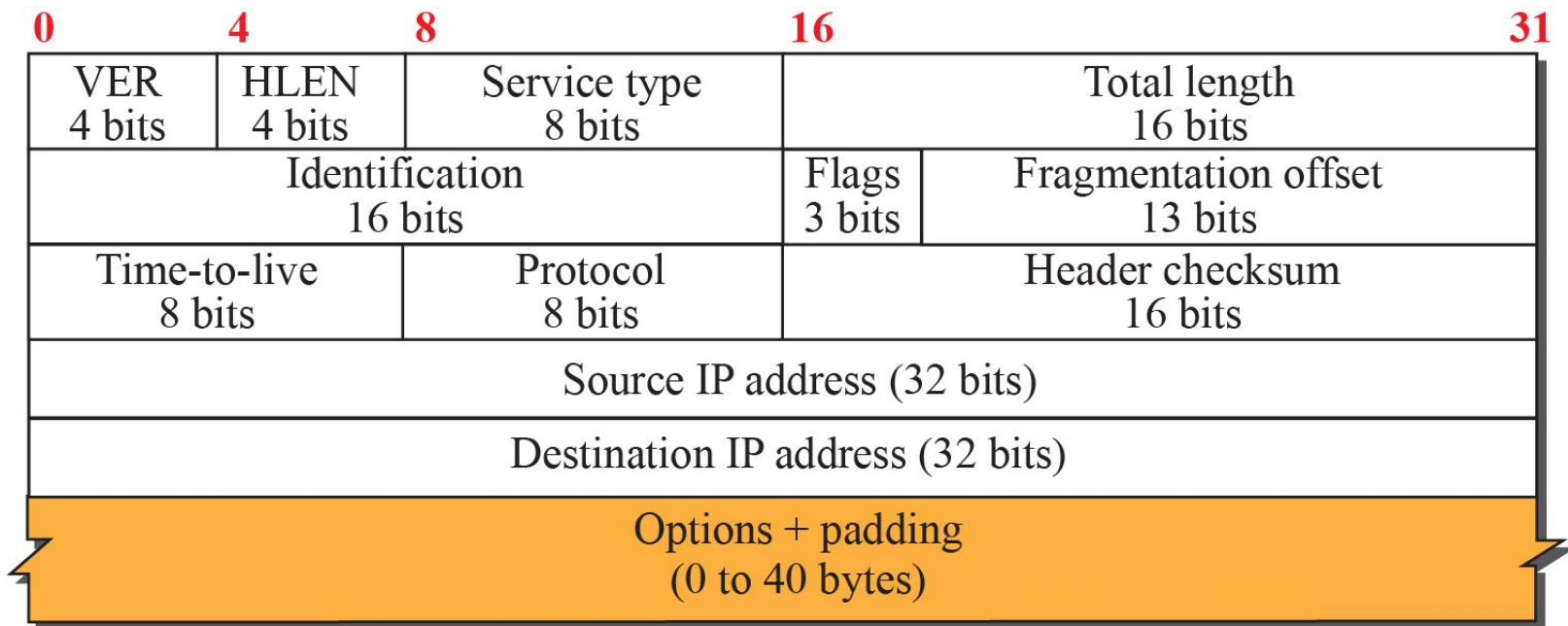
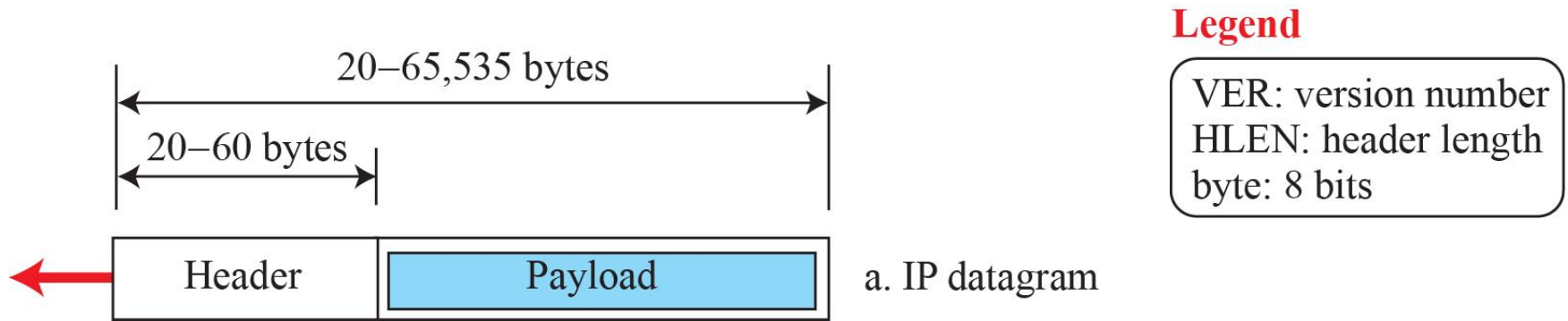




Note

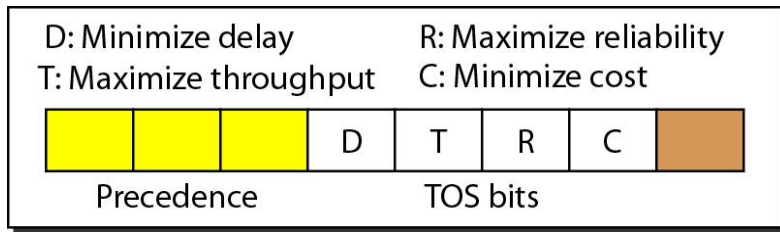
IP datagram

Figure 19.2: IP datagram

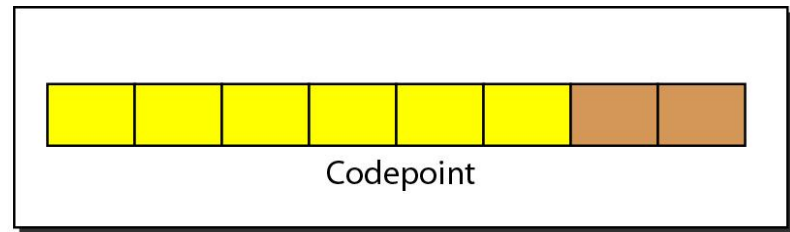


b. Header format

Figure 4.6 *Service type or differentiated services*



Service type



Differentiated services



Note

The precedence subfield was part of version 4, but never used.

Table 4.1 *Types of service*

<i>TOS Bits</i>	<i>Description</i>
0000	Normal (default)
0001	Minimize cost
0010	Maximize reliability
0100	Maximize throughput
1000	Minimize delay

Table 4.2 *Default types of service*

<i>Protocol</i>	<i>TOS Bits</i>	<i>Description</i>
ICMP	0000	Normal
BOOTP	0000	Normal
NNTP	0001	Minimize cost
IGP	0010	Maximize reliability
SNMP	0010	Maximize reliability
TELNET	1000	Minimize delay
FTP (data)	0100	Maximize throughput
FTP (control)	1000	Minimize delay
TFTP	1000	Minimize delay
SMTP (command)	1000	Minimize delay
SMTP (data)	0100	Maximize throughput
DNS (UDP query)	1000	Minimize delay
DNS (TCP query)	0000	Normal
DNS (zone)	0100	Maximize throughput

Figure 19.3: *Multiplexing and demultiplexing using the value of the protocol field*

ICMP: 01	UDP: 17
IGMP: 02	OSPF: 89
TCP: 06	

Some protocol values

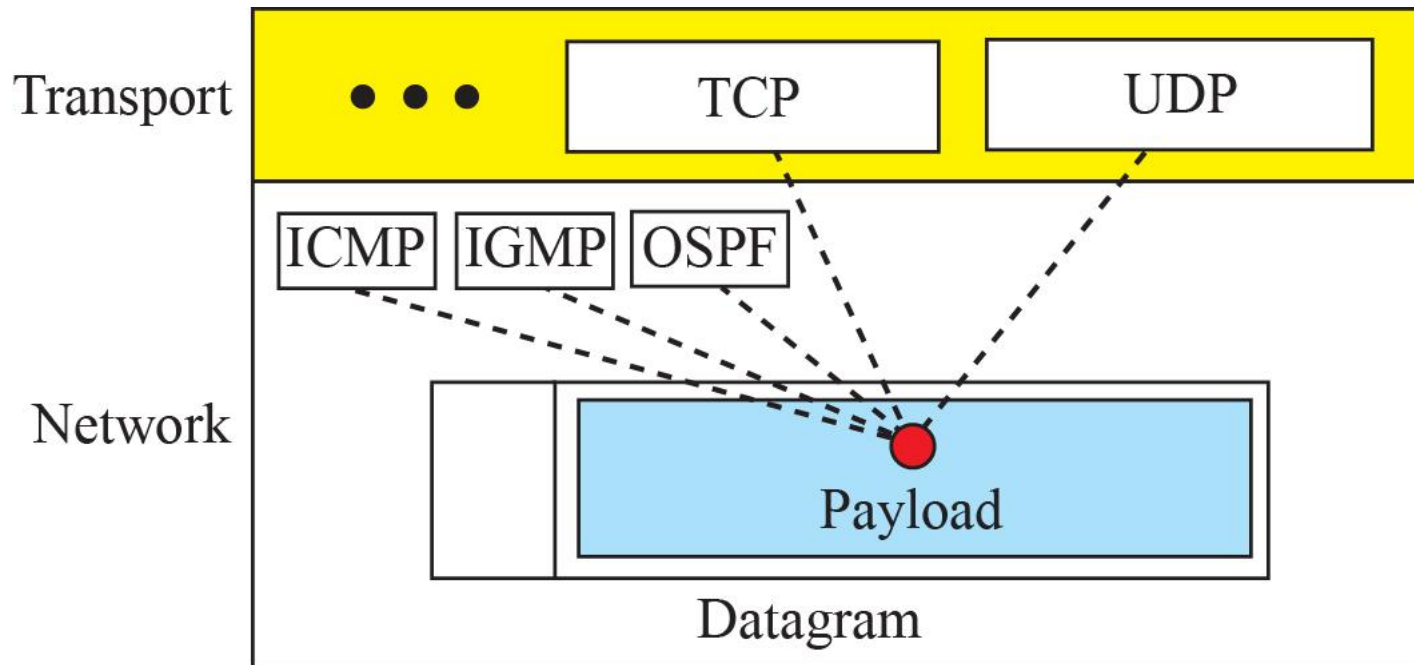
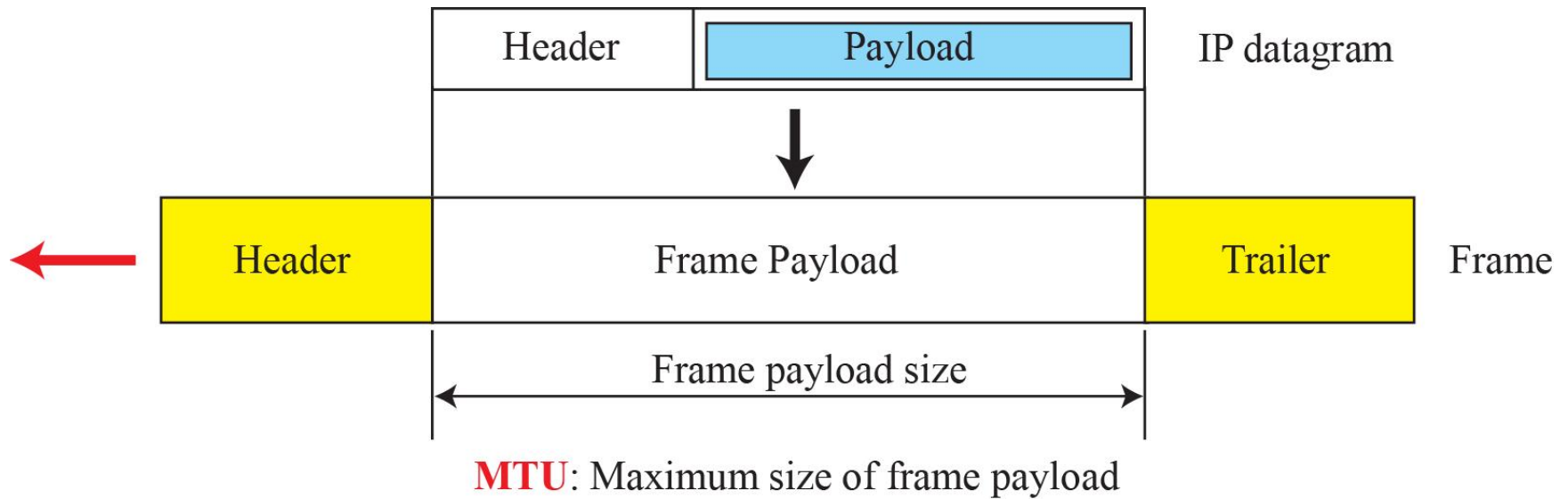


Figure 19.5: Maximum transfer unit (MTU)





Note

TCP datagram

Figure 6.16 *TCP segment format*

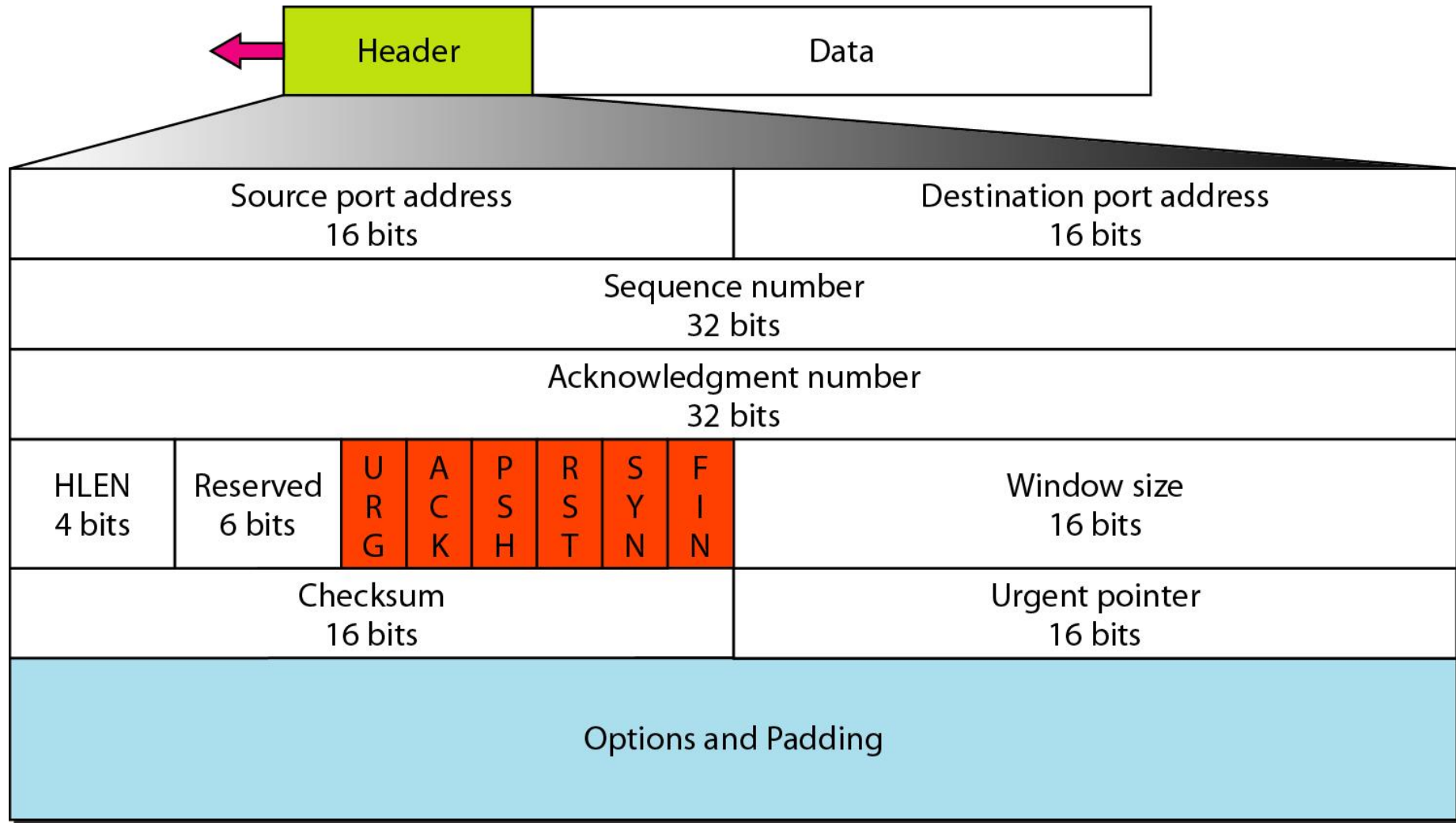


Figure 6.17 *Control field*

URG: Urgent pointer is valid
ACK: Acknowledgment is valid
PSH: Request for push

RST: Reset the connection
SYN: Synchronize sequence numbers
FIN: Terminate the connection



Table 6.3 *Description of flags in the control field*

<i>Flag</i>	<i>Description</i>
URG	The value of the urgent pointer field is valid.
ACK	The value of the acknowledgment field is valid.
PSH	Push the data.
RST	Reset the connection.
SYN	Synchronize sequence numbers during connection.
FIN	Terminate the connection.

2-3 5G Architecture

An NG-RAN node is either:

- *a gNB, providing NR user plane and control plane protocol terminations towards the UE;*
- *an ng-eNB, providing E-UTRA user plane and control plane protocol terminations towards the UE.*

The gNBs and ng-eNBs are interconnected with each other by means of the Xn interface. The gNBs and ng-eNBs are also connected by means of the NG interfaces to the 5GC, more specifically to the AMF (Access and Mobility Management Function) by means of the NG-C interface and to the UPF (User Plane Function) by means of the NG-U interface (see TS 23.501 [3]).

Topics discussed in this section:

Overall Architecture

Functional Split

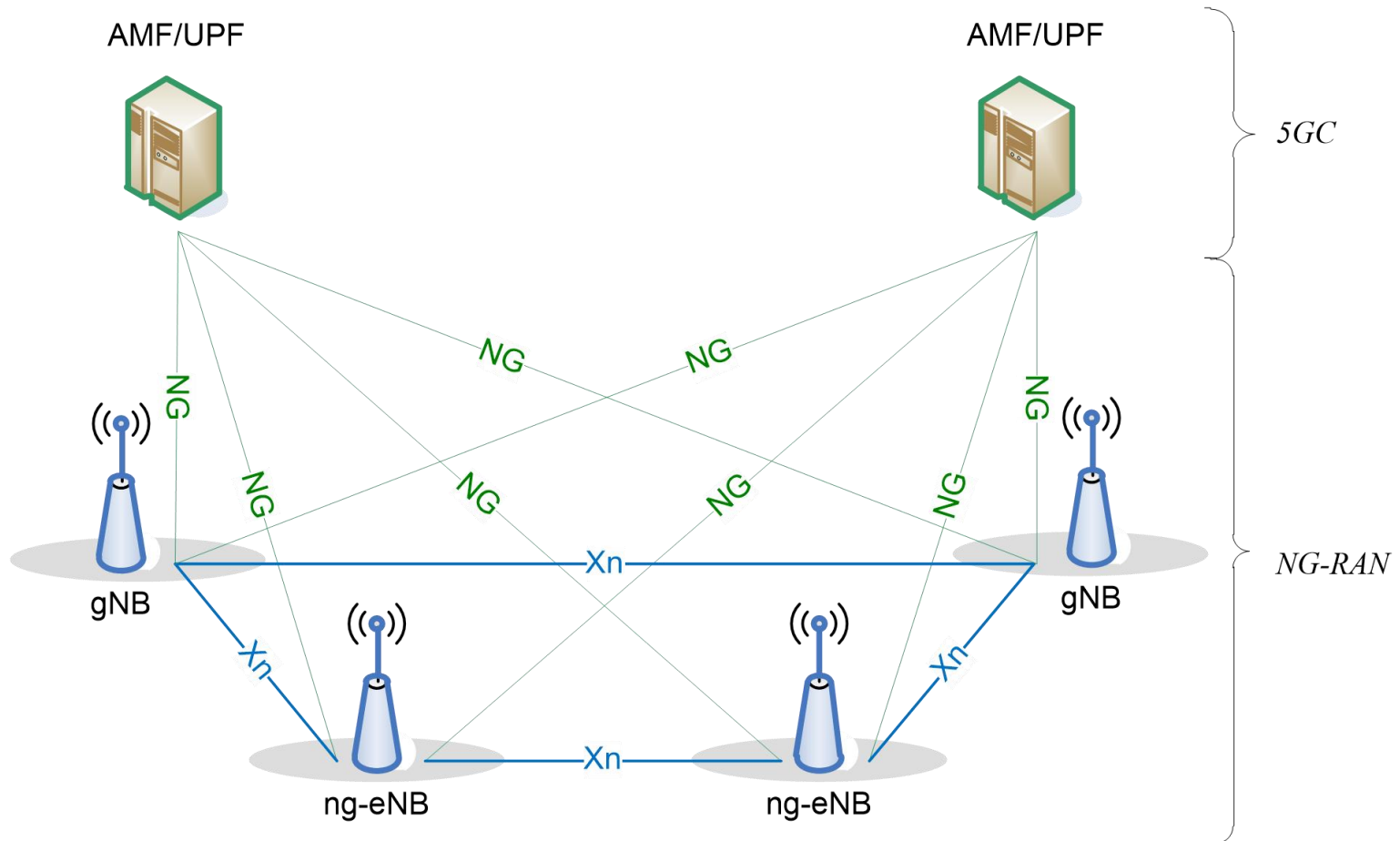
New Radio (NR) Protocol

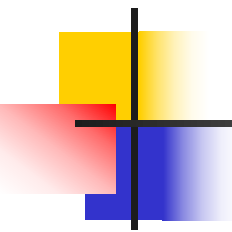
Overall Architecture (*TS38.300*)

AMF: Access and Mobility Management Function

UPF: User Plane Function

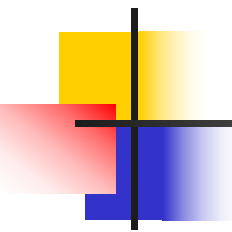
SMF: Session Management Function





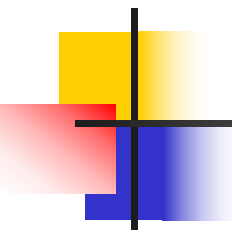
Functional Split – gNB & ng-eNB

- **Functions for Radio Resource Management:** Radio Bearer Control, Radio Admission Control, Connection Mobility Control, Dynamic allocation of resources to UEs in both uplink and downlink (scheduling);
 - IP and Ethernet header compression, uplink data decompression, encryption and integrity protection of data;
 - Selection of an AMF at UE attachment when no routing to an AMF can be determined from the information provided by the UE;
 - Routing of User Plane data towards UPF(s);
 - Routing of Control Plane information towards AMF;
 - **Connection setup and release;**
 - Scheduling and transmission of paging messages;
 - Scheduling and transmission of system broadcast information (originated from the AMF or OAM);
 - Measurement and measurement reporting configuration for mobility and scheduling;
 - Transport level packet marking in the uplink;
 - **Session Management;**
 - **Support of Network Slicing;**
 - **QoS Flow management and mapping to data radio bearers;**
 - Support of UEs in RRC_INACTIVE state;
 - Distribution function for NAS messages;
 - Radio access network sharing;
 - Dual Connectivity;
 - Tight interworking between NR and E-UTRA;
 - Maintain security and radio configuration for User Plane CIoT 5GS Optimisation, as defined in TS 23.501 [3] (ng-eNB only).
- NOTE 1: BL UE or UE in enhanced coverage is only supported by ng-eNB, see TS 36.300 [2].
- NOTE 2: NB-IoT UE is only supported by ng-eNB, see TS 36.300 [2].



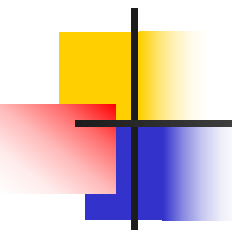
Functional Split – AMF

- NAS signalling termination;
- **NAS signalling security;**
- AS Security control;
- Inter CN node signalling for mobility between 3GPP access networks;
- Idle mode UE Reachability (including control and execution of paging retransmission);
- Registration Area management;
- Support of intra-system and inter-system mobility;
- **Access Authentication;**
- Access Authorization including check of roaming rights;
- **Mobility management control** (subscription and policies);
- **Support of Network Slicing;**
- SMF selection.
- Selection of CIoT 5GS optimisations;



Functional Split – UPF

- Anchor point for Intra-/Inter-RAT mobility (when applicable);
- External PDU session point of interconnect to Data Network;
- **Packet routing & forwarding;**
- Packet inspection and User plane part of Policy rule enforcement;
- Traffic usage reporting;
- Uplink classifier to support routing traffic flows to a data network;
- Branching point to support multi-homed PDU session;
- **QoS handling for user plane, e.g. packet filtering, gating, UL/DL rate enforcement;**
- Uplink Traffic verification (SDF to QoS flow mapping);
- Downlink packet buffering and downlink data notification triggering.



Functional Split – SMF

- **Session Management;**
- **UE IP address allocation and management;**
- Selection and control of UP function;
- Configures traffic steering at UPF to route traffic to proper destination;
- **Control part of policy enforcement and QoS;**
- Downlink Data Notification.

Functional Split

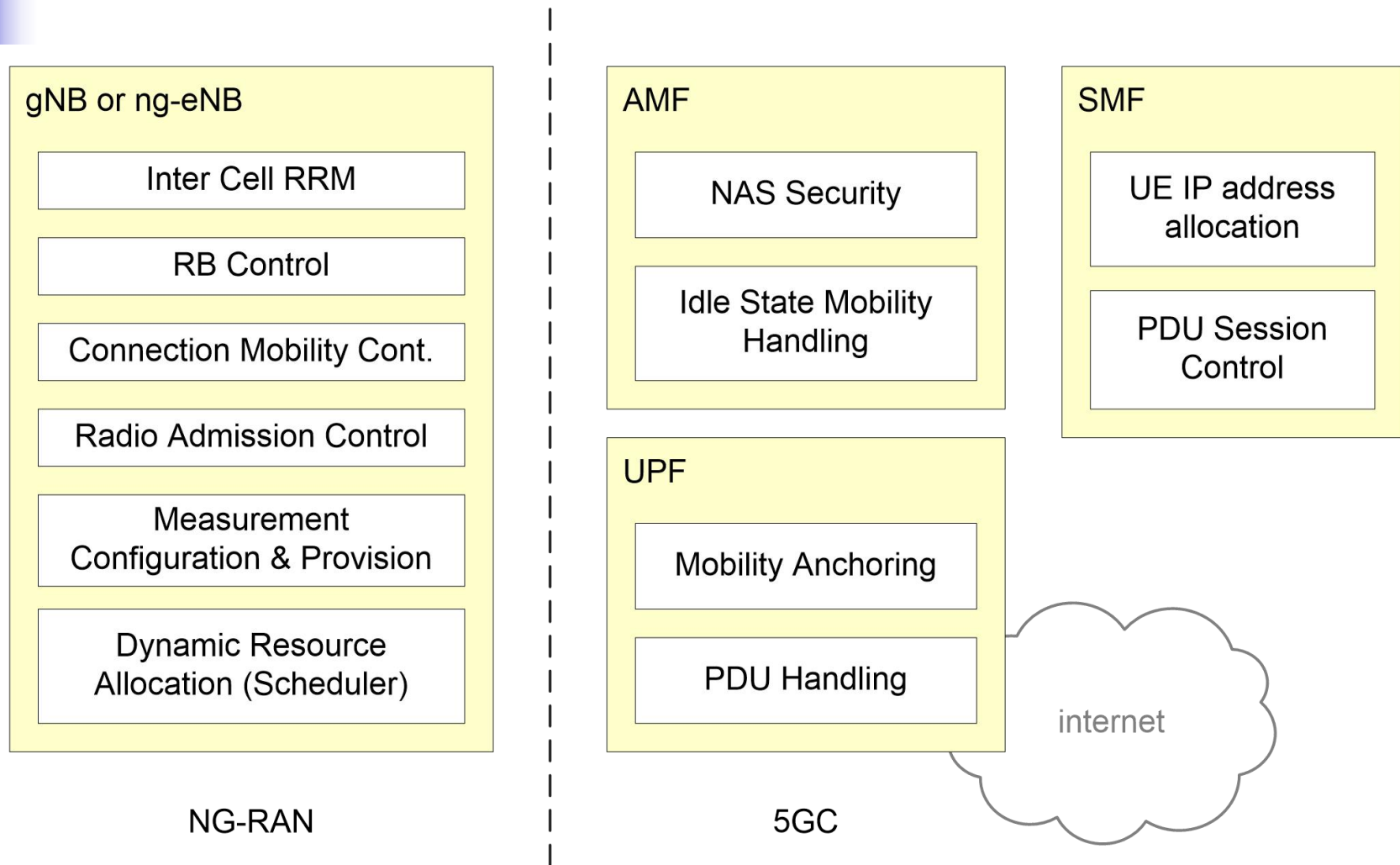
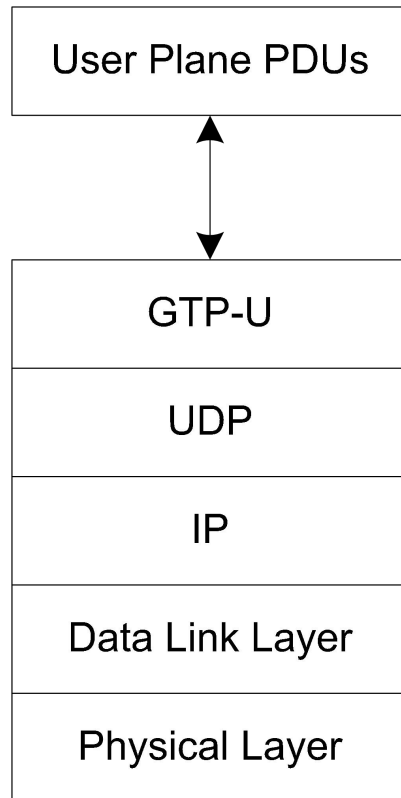


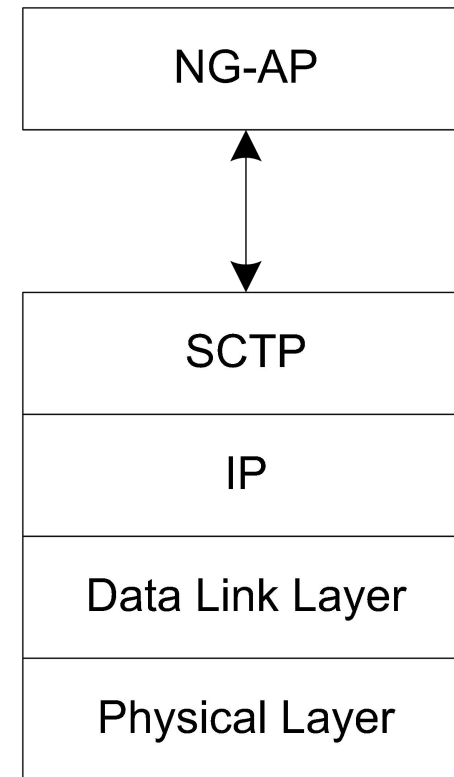
Figure 4.2-1: Functional Split between NG-RAN and 5GC

NG Interface Protocol Architecture

- 1. NG-U:** It provides non-guaranteed delivery of user plane PDUs between the NG-RAN node and the UPF.
- 2. NG-C:** The NG control plane interface is defined between the NG-RAN node and the AMF.

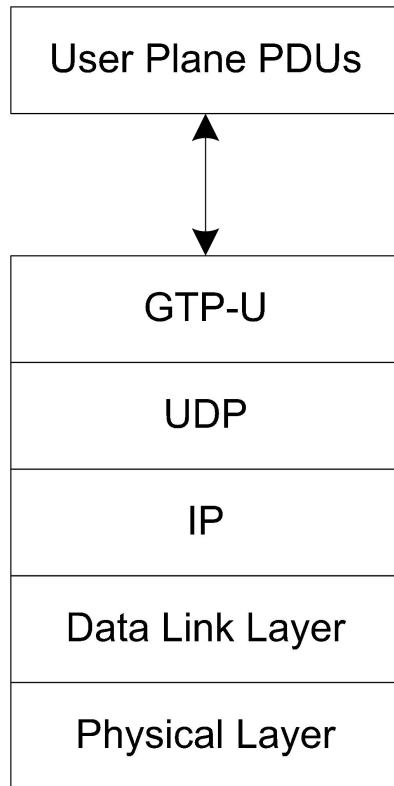


NG-U Protocol Stack

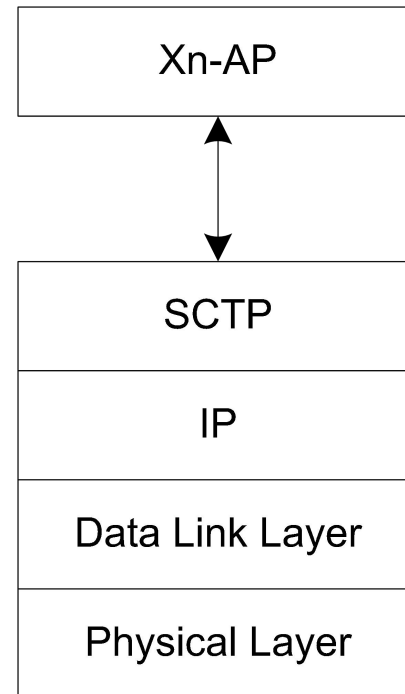


NG-C Protocol Stack

Xn Interface Protocol Architecture



Xn-U Protocol Stack



Xn-C Protocol Stack

Radio Protocol Architecture – Control Plane

NAS: Non-Access Stratum

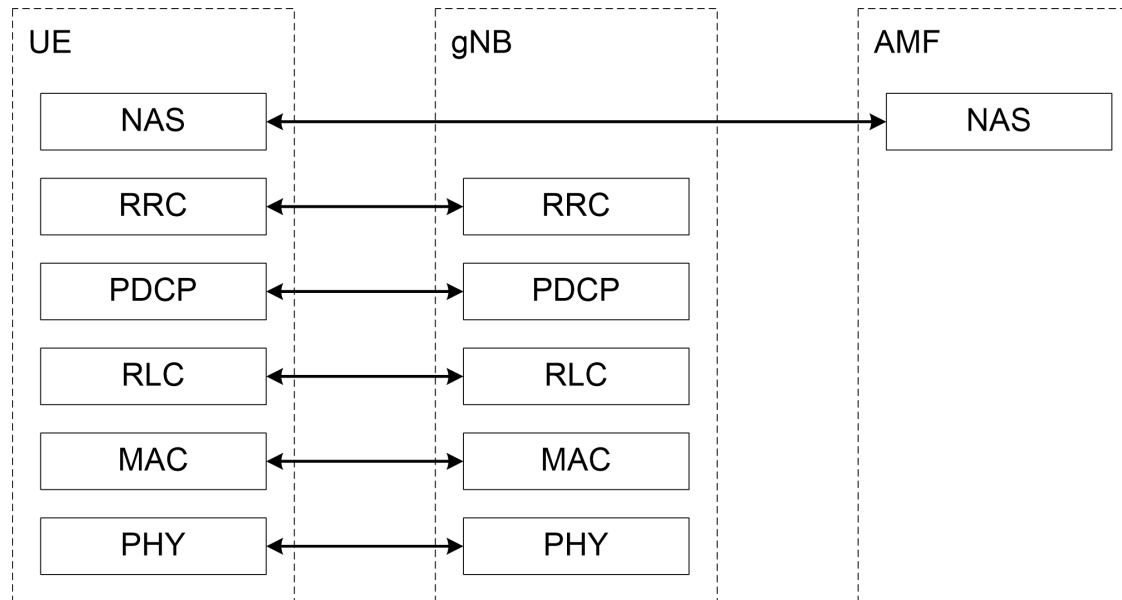
RRC: Radio Resource Control

PDCP: Packet Data Convergence Protocol

RLC: Radio Link Control

MAC: Medium Access Control

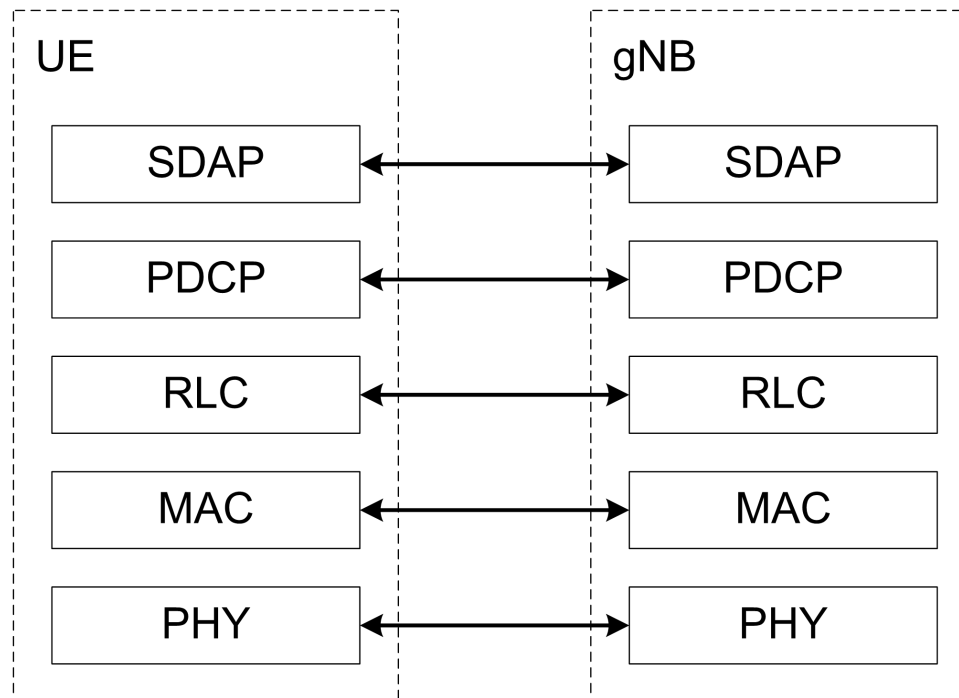
PHY: Physical Layer



Control Plane Protocol Stack

Radio Protocol Architecture – User Plane

SDAP: Service Data Adaptation Protocol
PDCCP: Packet Data Convergence Protocol
RLC: Radio Link Control
MAC: Medium Access Control
PHY: Physical Layer



User Plane Protocol Stack

Part I

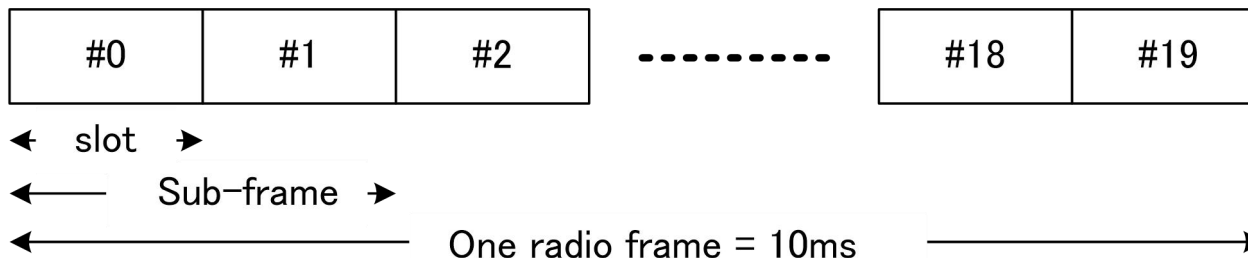
NR Physical Layer

LTE FDD - Frame Structure Type 1

Uplink and downlink transmissions are separated in the frequency domain.
Frame structure Type 1 is illustrated in the following Figure.

Each 10 ms radio frame is divided into ten equally sized sub-frames.
Each sub-frame consists of two equally sized slots.

For FDD, in each 10 ms interval,
10 subframes are available for downlink transmission,
and 10 subframes are available for uplink transmissions.



FDD - Frame structure type 1

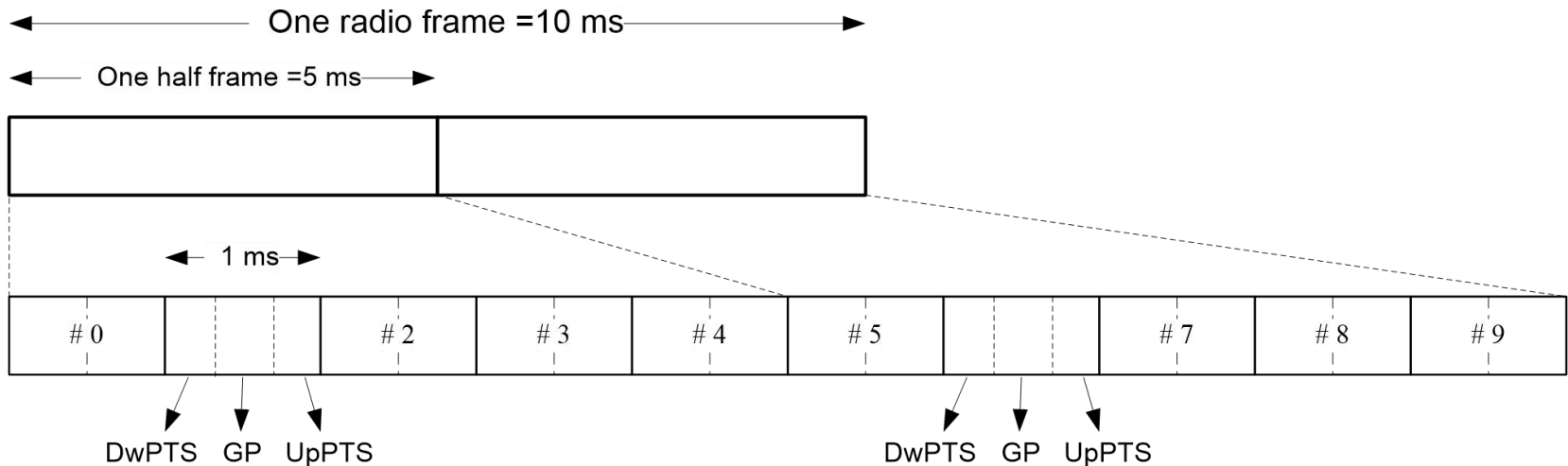
LTE TDD - Frame Structure Type2

Frame structure Type 2 is illustrated in the following Figure 5.1-2.

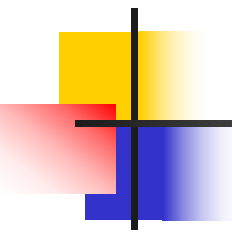
Each 10 ms radio frame consists of two half-frames of 5 ms each.

Each half-frame consists of eight slots of length 0.5 ms and three special fields: DwPTS, GP and UpPTS.

The total length of DwPTS, GP and UpPTS being equal to 1ms. Both 5ms and 10ms switch-point periodicity are supported.



TDD - Frame structure type 2 (for 5ms switch-point periodicity)

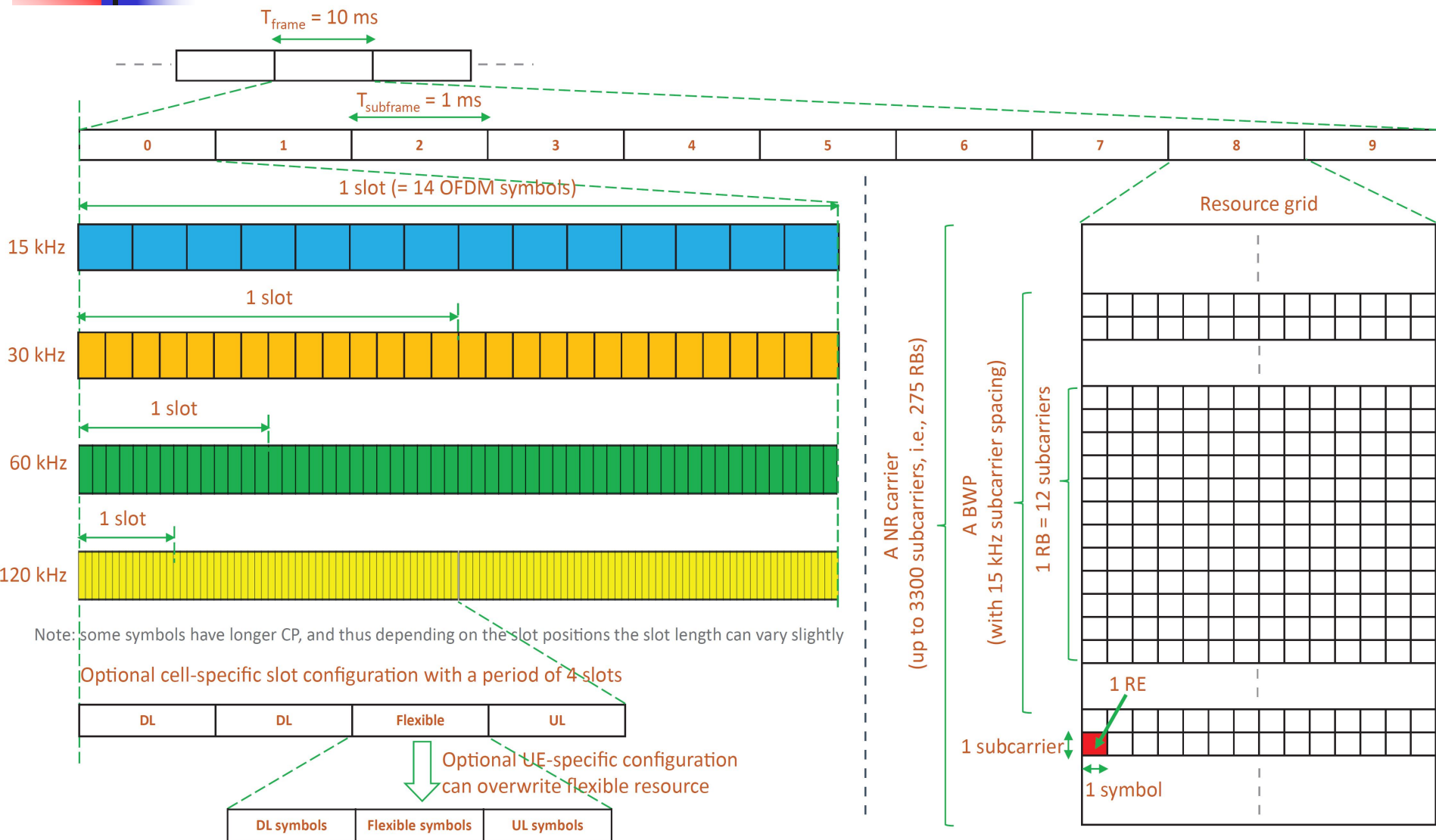


LTE TDD Predefined Configurations

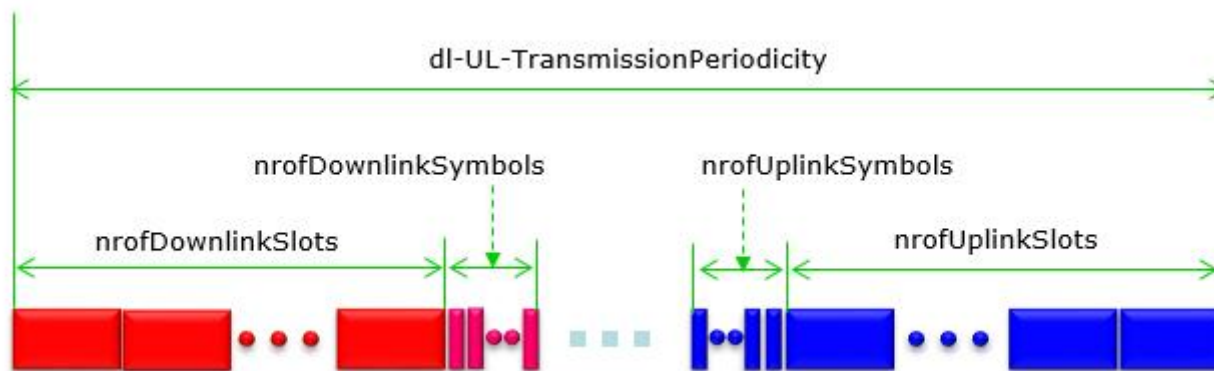
Uplink-downlink allocations.

Config uration	Switch- point periodi city	Subframe number									
		0	1	2	3	4	5	6	7	8	9
0	5 ms	D	S	U	U	U	D	S	U	U	U
1	5 ms	D	S	U	U	D	D	S	U	U	D
2	5 ms	D	S	U	D	D	D	S	U	D	D
3	10 ms	D	S	U	U	U	D	D	D	D	D
4	10 ms	D	S	U	U	D	D	D	D	D	D
5	10 ms	D	S	U	D	D	D	D	D	D	D
6	5 ms	D	S	U	U	U	D	S	U	U	D

NR Radio Frame Structure



NR TDD UL/DL Flexible Configuration



These parameters are defined in 38.331 v15.3.0 as follows :

```

TDD-UL-DL-ConfigCommon ::= SEQUENCE {
    referenceSubcarrierSpacing
    pattern1
    pattern2
    ...
}
TDD-UL-DL-Pattern ::= SEQUENCE {
    dl-UL-TransmissionPeriodicity
    nrofDownlinkSlots
    nrofDownlinkSymbols
    nrofUplinkSlots
    nrofUplinkSymbols
    ...,
    [[
        dl-UL-TransmissionPeriodicity-v1530 ENUMERATED {ms3, ms4} OPTIONAL -- Need R
    ]]
}
maxNrofSlots INTEGER ::= 320 // Maximum number of slots in a 10 ms period
maxNrofSymbols-1 INTEGER ::= 13 // Maximum index identifying a symbol within a slot
                                   (14 symbols, indexed from 0..13)
    
```

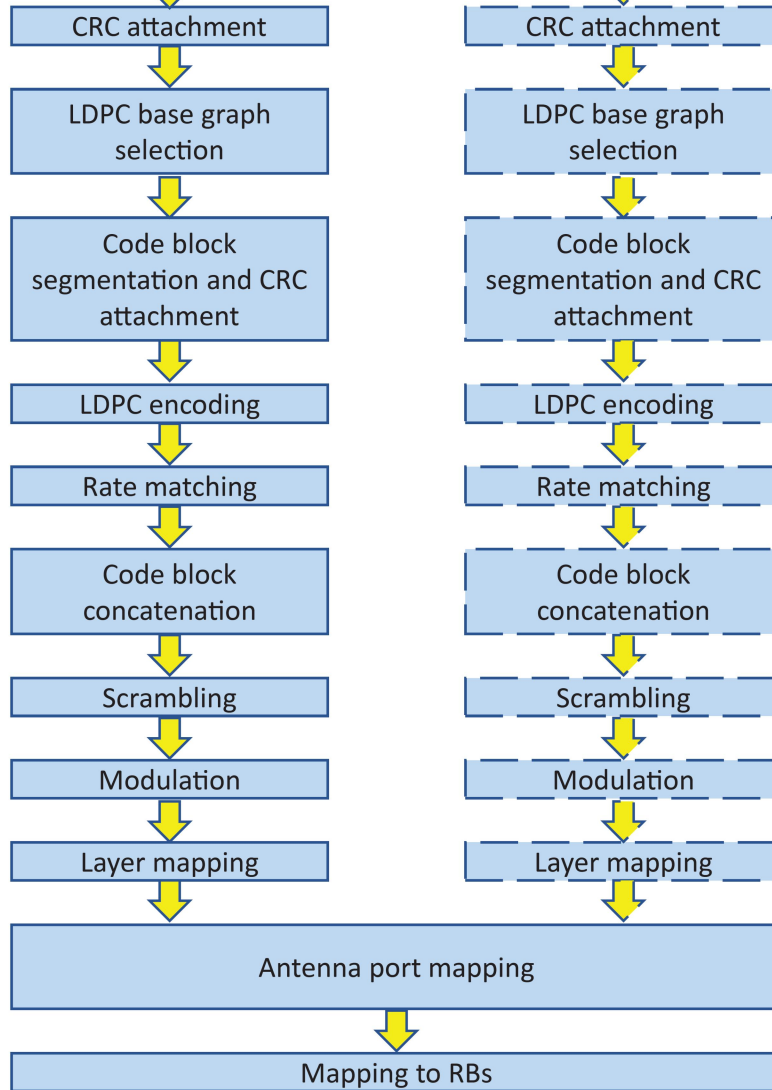
SubcarrierSpacing,
TDD-UL-DL-Pattern,
TDD-UL-DL-Pattern OPTIONAL,
...

ENUMERATED {ms0p5, ms0p625, ms1, ms1p25, ms2, ms2p5, ms5, ms10},
INTEGER (0..maxNrofSlots),
INTEGER (0..maxNrofSymbols-1),
INTEGER (0..maxNrofSlots),
INTEGER (0..maxNrofSymbols-1),

Physical Layer Processing for PDSCH & PUSCH

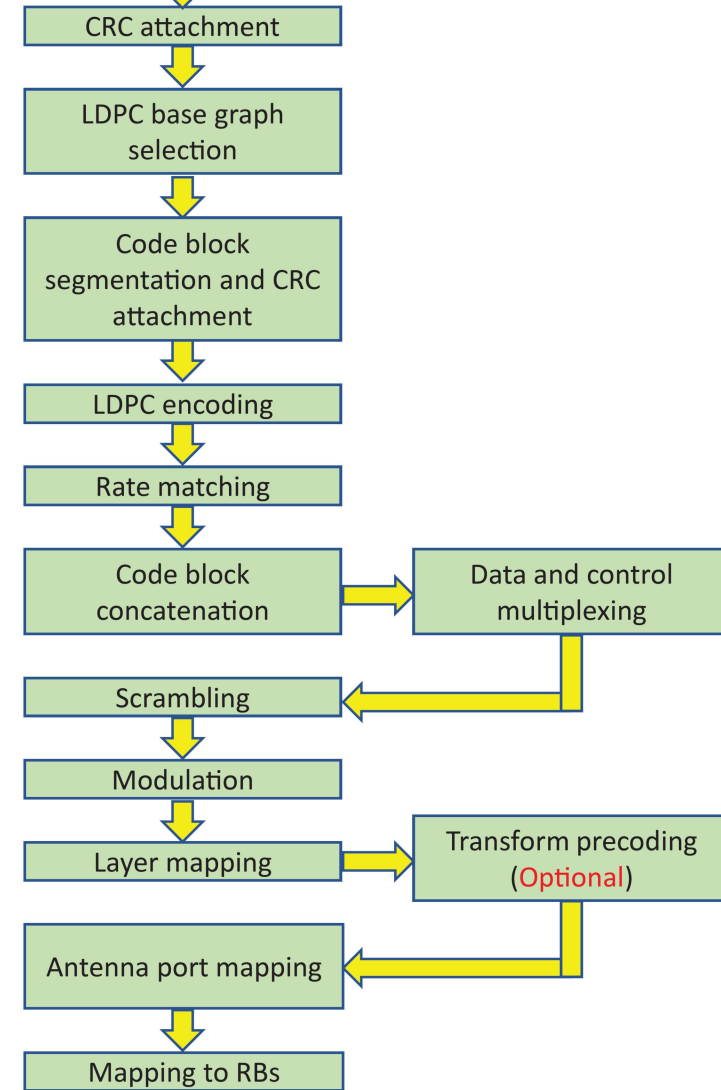
PDSCH physical-layer processing

A transport block



PUSCH physical-layer processing

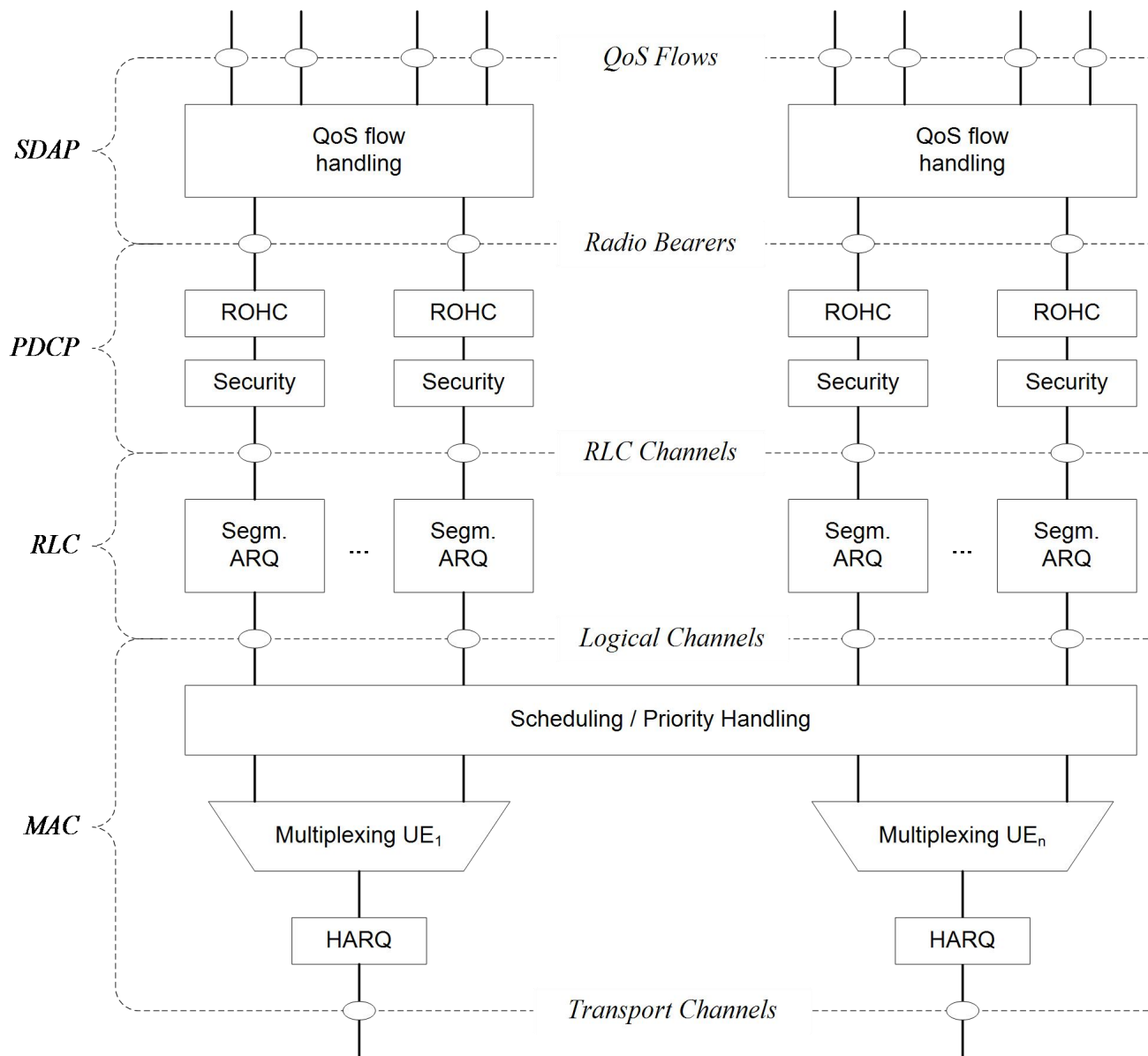
A transport block



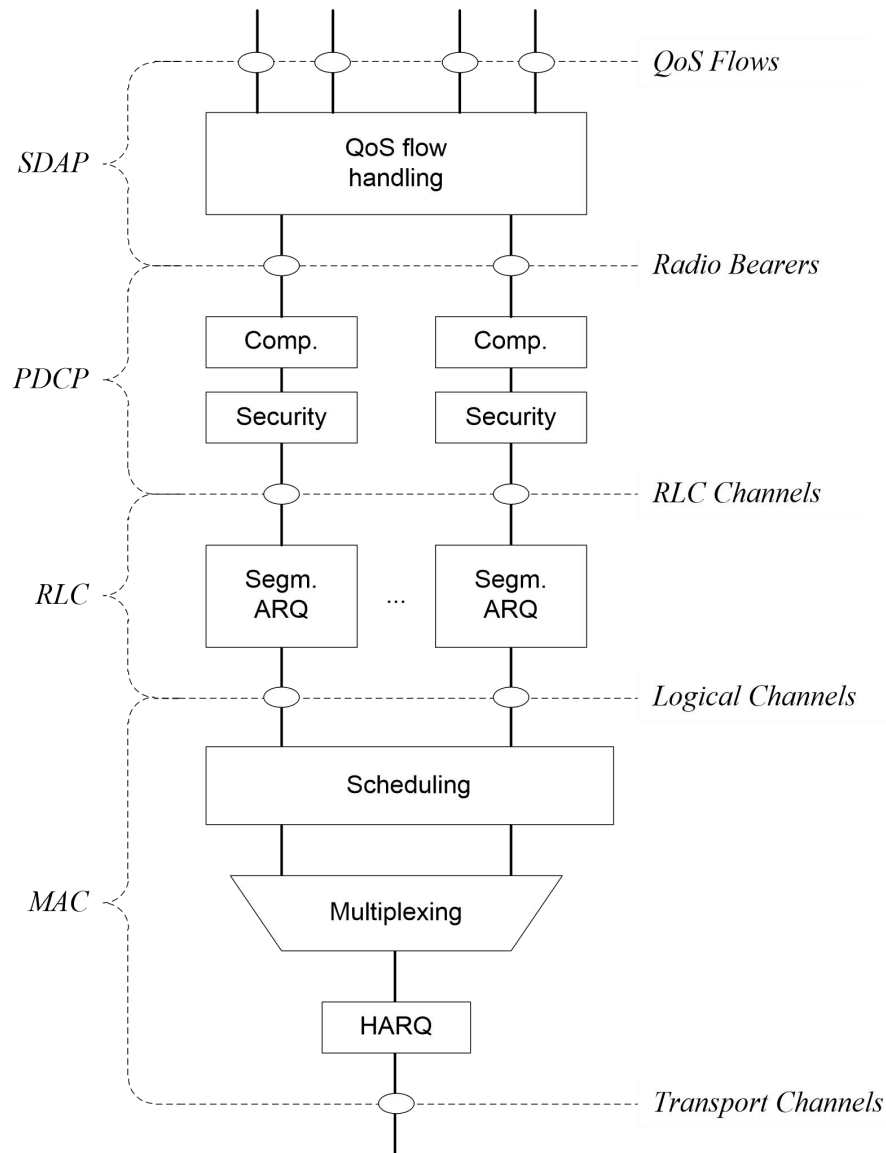
Part II

NR Layer 2

NR Layer 2 Structure for DL

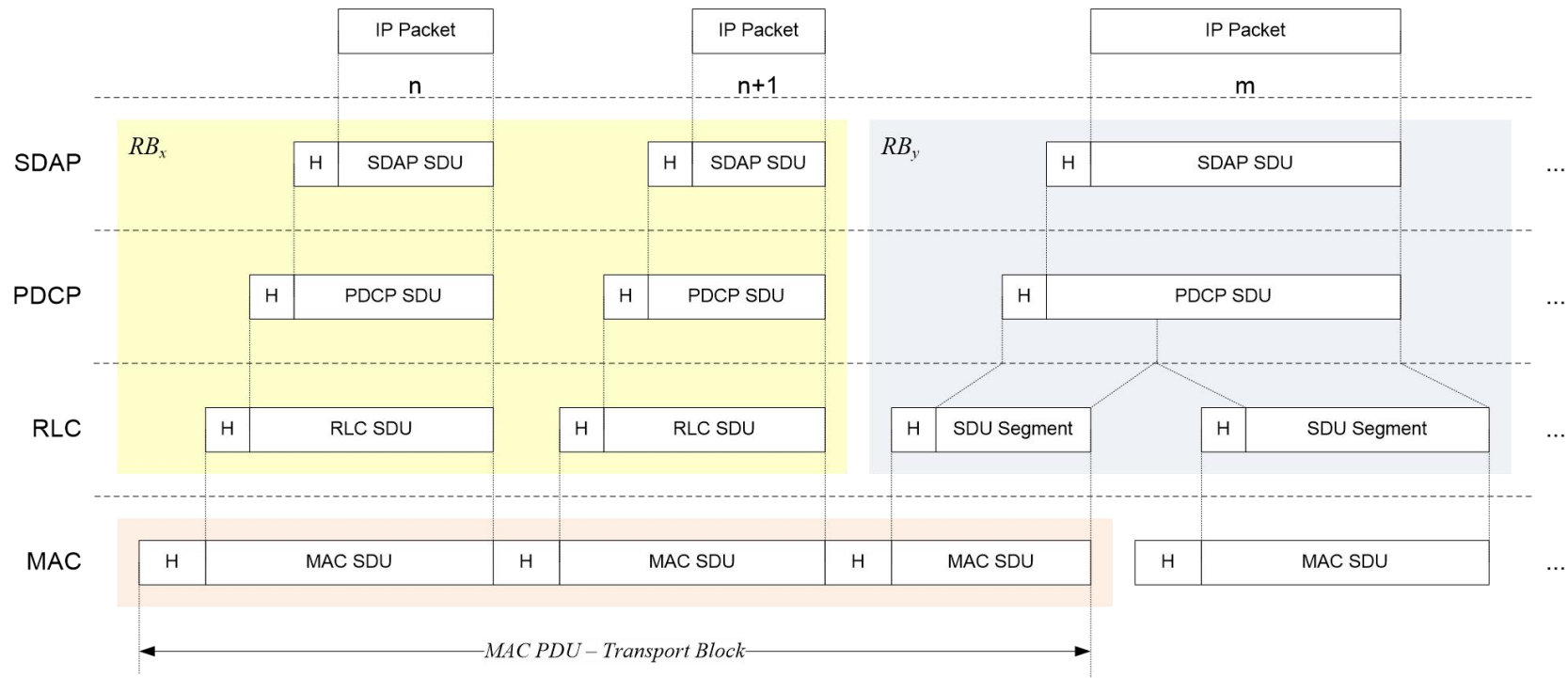


NR Layer 2 Structure for UL

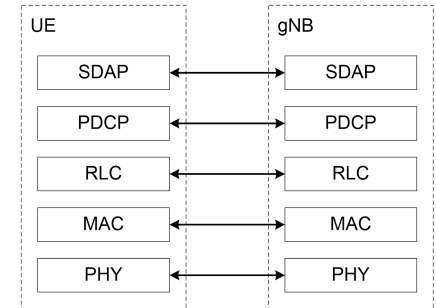


An Example of User Plane Layer 2 Data Flow

As shown in the following Figure, where a transport block is generated by MAC by concatenating two RLC PDUs from RB_x and one RLC PDU from RB_y . The two RLC PDUs from RB_x each corresponds to one IP packet (n and $n+1$) while the RLC PDU from RB_y is a segment of an IP packet (m).



SDAP Sublayer: Services and Functions



User Plane Protocol Stack

The main services and functions of SDAP include:

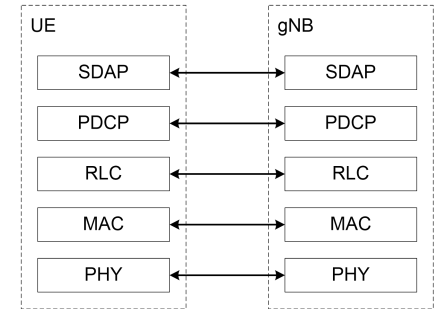
- Mapping between a QoS flow and a data radio bearer;
- Marking QoS flow ID (QFI) in both DL and UL packets.

A single protocol entity of SDAP is configured for each individual PDU session.

PDCP Sublayer: Services and Functions

The main services and functions of PDCP include:

- **Transfer of data (user plane or control plane);**
- **Maintenance of PDCP SNs;**
- **Ciphering and deciphering;**
- **Integrity protection and integrity verification;**
- **Timer based SDU discard;**
- **For split bearers, routing;**
- **Duplication;**
- **Reordering and in-order delivery;**
- **Out-of-order delivery;**
- **Duplicate discarding.**
- **Header compression and decompression using the ROHC protocol;**
- **Header compression and decompression using EHC protocol;**
- **Compression and decompression of uplink PDCP SDUs;**

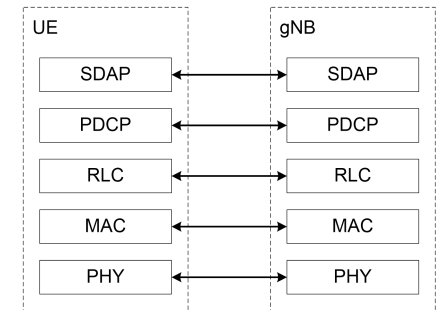


User Plane Protocol Stack

RLC Sublayer: Services and Functions

The RLC sublayer supports three transmission modes:

- **Transparent Mode (TM);**
- **Unacknowledged Mode (UM);**
- **Acknowledged Mode (AM).**

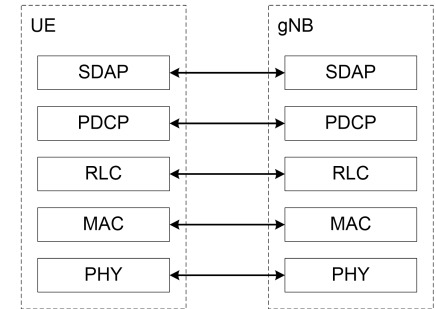


User Plane Protocol Stack

The main services and functions of the RLC sublayer depend on the transmission mode and include:

- Transfer of upper layer PDUs;
- Sequence numbering independent of the one in PDCP (UM and AM);
- Error Correction through ARQ (AM only);
- Segmentation (AM and UM) and re-segmentation (AM only) of RLC SDUs;
- Reassembly of SDU (AM and UM);
- Duplicate Detection (AM only);
- RLC SDU discard (AM and UM);
- RLC re-establishment;
- Protocol error detection (AM only).

MAC Sublayer: Services and Functions



User Plane Protocol Stack

The main services and functions of MAC include:

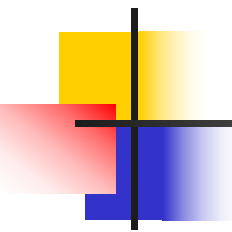
- Mapping between logical channels and transport channels;
- Multiplexing/demultiplexing of MAC SDUs belonging to one or different logical channels into/from transport blocks (TB) delivered to/from the physical layer on transport channels;
- Scheduling information reporting;
- Error correction through HARQ (one HARQ entity per cell in case of CA);
- Priority handling between UEs by means of dynamic scheduling;
- Priority handling between logical channels of one UE by means of logical channel prioritisation;
- Priority handling between overlapping resources of one UE;
- Padding.

A single MAC entity can support multiple numerologies, transmission timings and cells. Mapping restrictions in logical channel prioritization control which numerology(ies), cell(s), and transmission timing(s) a logical channel can use.



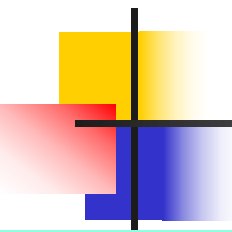
Part III

NR Radio Resource Control (RRC)



RRC: Services and Functions

- Broadcast of System Information related to AS and NAS;
- **Paging initiated by 5GC or NG-RAN;**
- Establishment, maintenance and release of an RRC connection between the UE and NG-RAN ;
- Security functions including key management;
- Establishment, configuration, maintenance and release of Signalling Radio Bearers (SRBs) and Data Radio Bearers (DRBs);
- **Mobility functions including:** Handover and context transfer; UE cell selection and reselection and control of cell selection and reselection; Inter-RAT mobility.
- **QoS management functions;**
- **UE measurement reporting and control of the reporting;**
- Detection of and recovery from radio link failure;
- NAS message transfer to/from NAS from/to UE.



RRC Supports Protocol States

- RRC_INACTIVE:

- PLMN selection;
- Broadcast of system information;
- Cell re-selection mobility;
- Paging is initiated by NG-RAN (RAN paging);
- RAN-based notification area (RNA) is managed by NG- RAN;
- DRX for RAN paging configured by NG-RAN;
- 5GC - NG-RAN connection (both C/U-planes) is established for UE;
- The UE Inactive AS context is stored in NG-RAN and the UE;
- NG-RAN knows the RNA which the UE belongs to;
- Transfer of unicast data and/or signalling to/from the UE over radio bearers configured for SDT.

- RRC_CONNECTED:

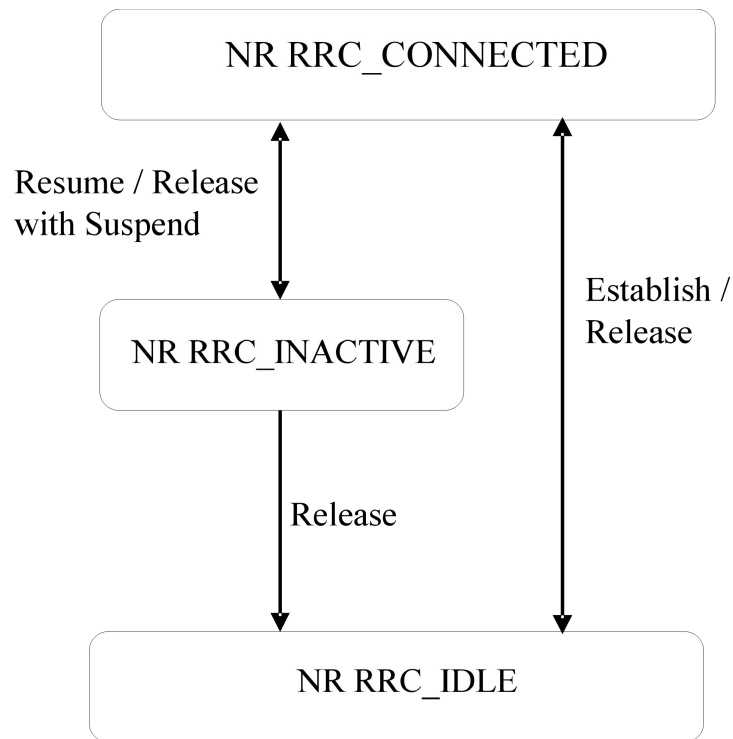
- 5GC - NG-RAN connection (both C/U-planes) is established for UE;
- The UE AS context is stored in NG-RAN and the UE;
- NG-RAN knows the cell which the UE belongs to;
- Transfer of unicast data to/from the UE;
- Network controlled mobility including measurements.

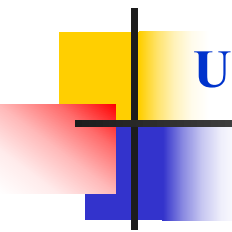
- RRC_IDLE:

- PLMN selection;
- Broadcast of system information;
- Cell re-selection mobility;
- Paging for mobile terminated data is initiated by 5GC;
- DRX for CN paging configured by NAS.

UE State Machine and State Transitions in NR (TS38.331)

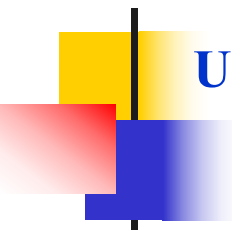
A UE is either in **RRC_CONNECTED** state or in **RRC_INACTIVE** state when an RRC connection has been established. If this is not the case, i.e. no RRC connection is established, the UE is in **RRC_IDLE** state.





UE at RRC IDLE State (TS38.331)

- A UE specific DRX may be configured by upper layers;
- At lower layers, the UE may be configured with a DRX for PTM transmission of MBS broadcast;
- UE controlled mobility based on network configuration;
- **The UE:**
 - **Monitors Short Messages transmitted with P-RNTI over DCI;**
 - **Monitors a Paging channel for CN paging using 5G-S-TMSI, except if the UE is acting as a L2 U2N Remote UE;**
 - **If configured by upper layers for MBS multicast reception, monitors a Paging channel for CN paging using TMGI;**
 - **Performs neighbouring cell measurements and cell (re-)selection;**
 - **Acquires system information and can send SI request (if configured);**
 - ...



UE at INACTIVE State (TS38.331)

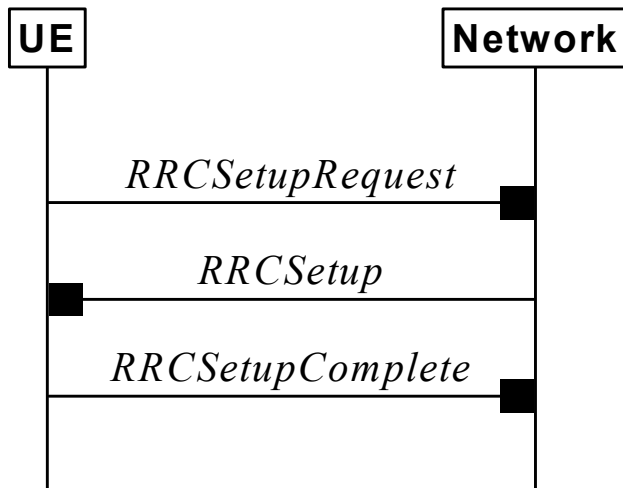
- A UE specific DRX may be configured by upper layers or by RRC layer;
- UE controlled mobility based on network configuration;
- The UE stores the UE Inactive AS context;
- A RAN-based notification area is configured by RRC layer;
- ...

The UE:

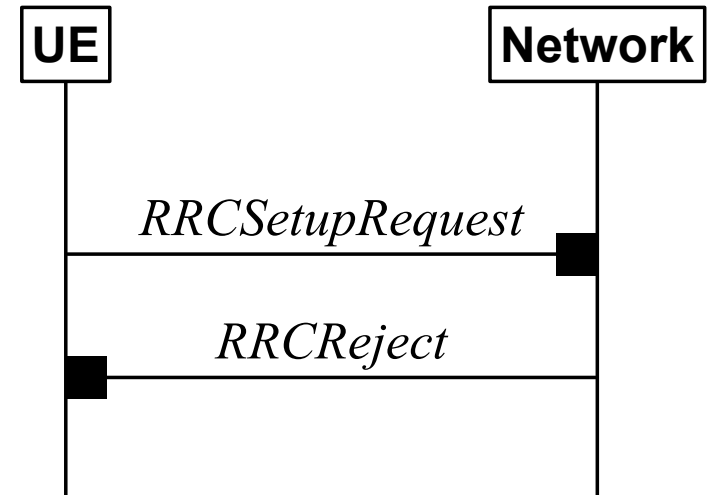
- **Monitors Short Messages transmitted with P-RNTI over DCI (see clause 6.5);**
- **During SDT procedure, monitors control channels associated with the shared data channel to determine if data is scheduled for it;**
- **Performs neighbouring cell measurements and cell (re-)selection;**
- ...

- The UE stores the AS context;
- Transfer of unicast data to/from UE;
- Transfer of MBS multicast data to UE;
- At lower layers, the UE may be configured with a UE specific DRX;
- Network controlled mobility within NR, to/from E-UTRA;
- ...
- **The UE:**
 - **Monitors Short Messages transmitted with P-RNTI over DCI, if configured;**
 - **Provides channel quality and feedback information;**
 - **Performs neighbouring cell measurements and measurement reporting;**
 - **Acquires system information;**
 - ...

RRC Connection Establishment (TS38.331)

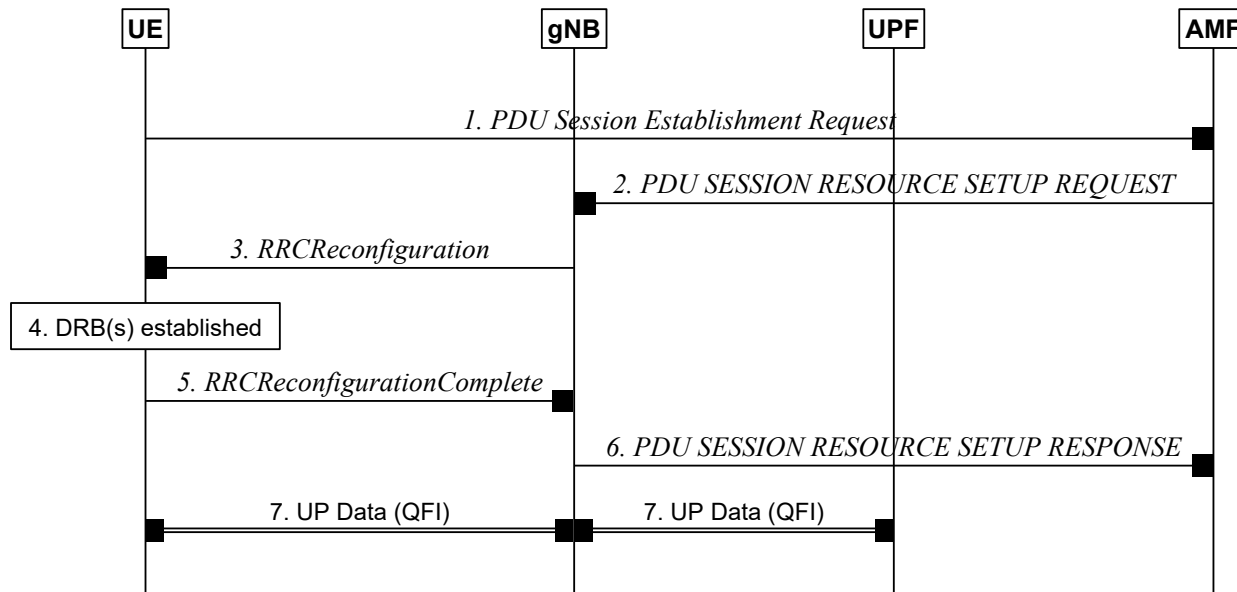


RRC connection establishment, successful



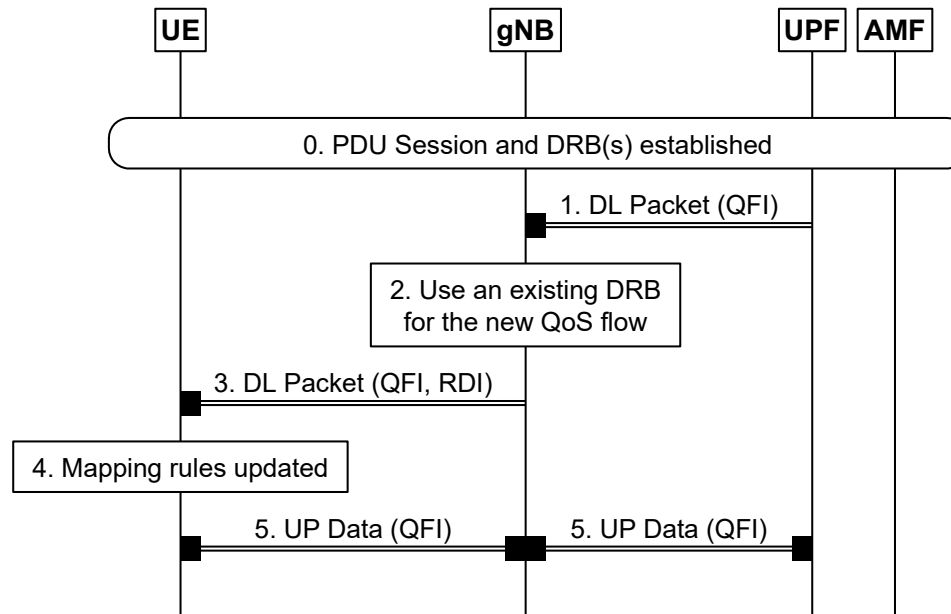
RRC connection establishment, network reject

PDU Session Establishment (*TS38.300*)



1. UE requests a PDU session establishment to AMF.
2. AMF sends a PDU SESSION RESOURCE SETUP REQUEST message to gNB, which includes the NAS message to be sent to the UE with NAS QoS related information.
3. gNB sends an *RRCReconfiguration* message to UE including the configuration of at least one DRB and the NAS message received at Step 2.
4. UE establishes the **Data Radio Bearers (DRBs)** for the new PDU session and creates the QFI to DRB mapping rules.
5. UE sends an *RRCReconfiguration Complete* message to gNB.
6. gNB sends a PDU SESSION RESOURCE SETUP RESPONSE message to AMF.
7. User Plane Data can then be exchanged between UE and gNB over DRB(s) according to the mapping rules and between UPF and gNB over the tunnel for the PDU session. QFI marking over Uu is optional (see clause 12) while QFI marking over NG-U is always present.

DL data with new QFI sent over existing DRB



0. PDU session and DRB(s) have been already established.

1. gNB receives a downlink packet with a new QFI from UPF.

2. gNB decides to send the new QoS flow over an existing DRB.

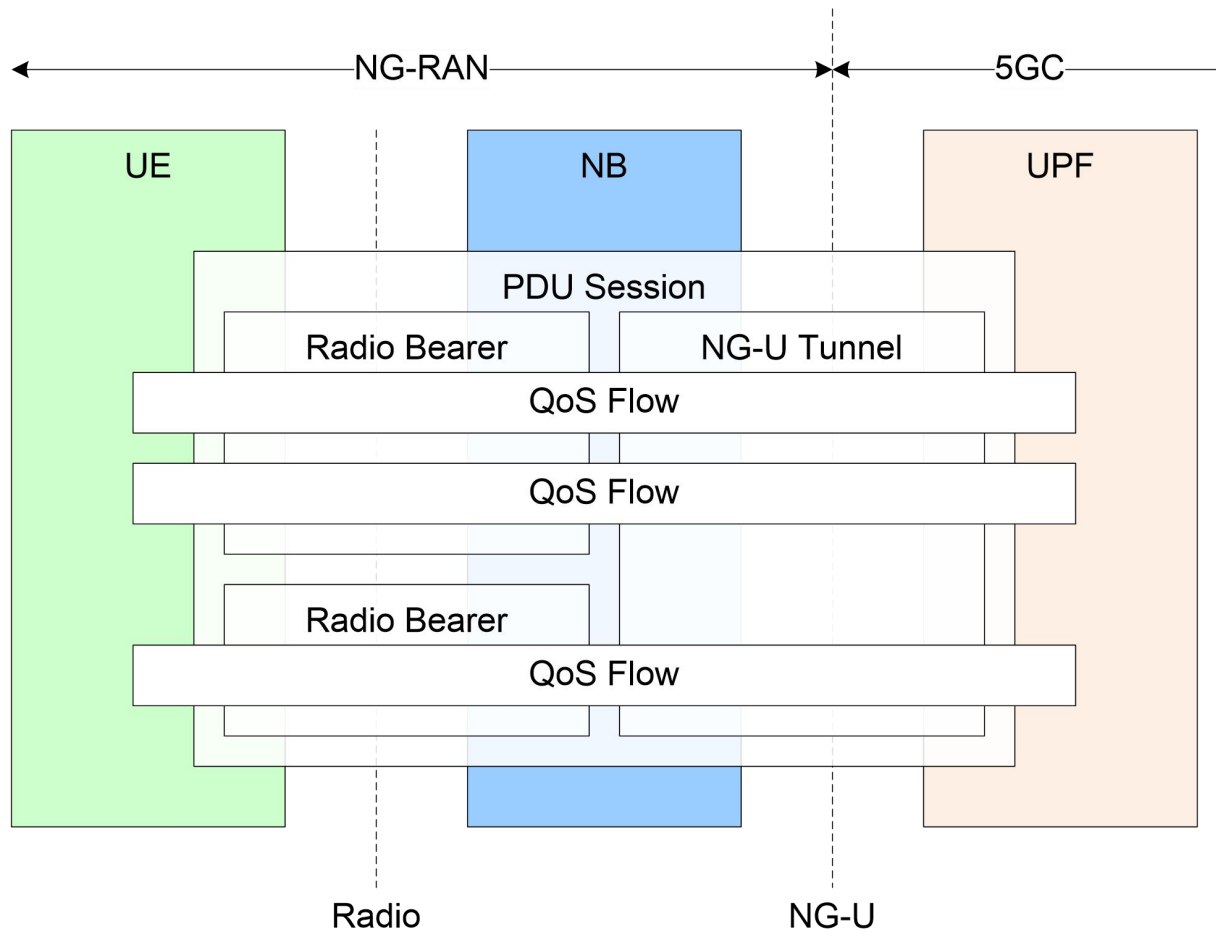
NOTE: If gNB decides to send it over a new DRB, it needs to establish the DRB first.

3. gNB sends the DL packet over the selected DRB with the new QFI and RDI set in the SDAP header.

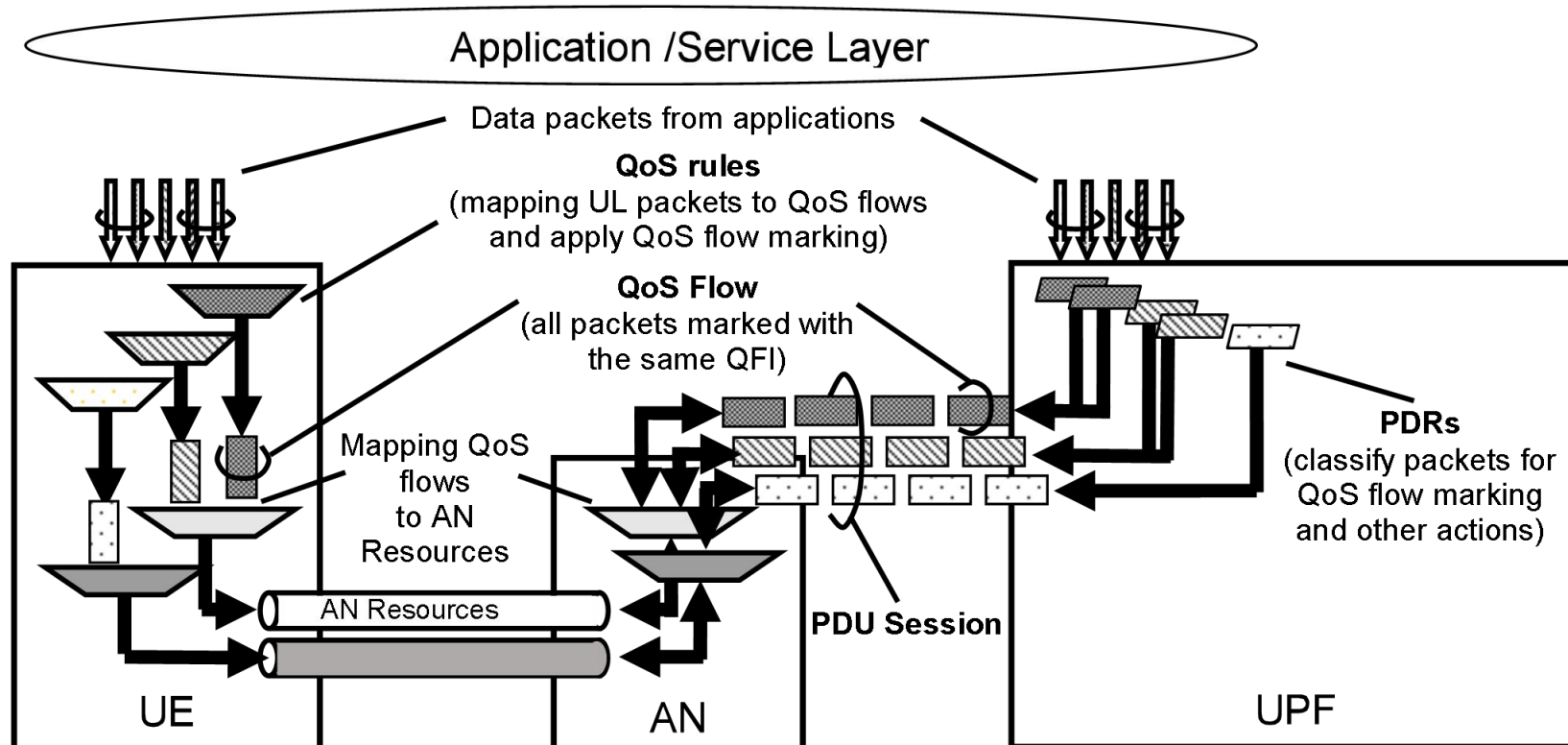
4. UE identifies the QFI and RDI in the received DL packet and the DRB on which the packet was received. The AS mapping rules are then updated accordingly.

5. User Plane Data for the new QoS flow can then be exchanged between UE and gNB over the DRB according to the updated mapping rules and between UPF and gNB over the tunnel for the PDU session.

QoS Architecture (TS38.300)



QoS Flows and mapping to AN Resources (TS23.501)



A QoS Flow may either be **guaranteed flow bit rate (GBR QoS Flows)** and **QoS Flows that do not require guaranteed flow bit rate (Non-GBR QoS Flows)** depending on its QoS profile. The QoS profile of a QoS Flow is sent to the (R)AN and it contains QoS parameters as described below:

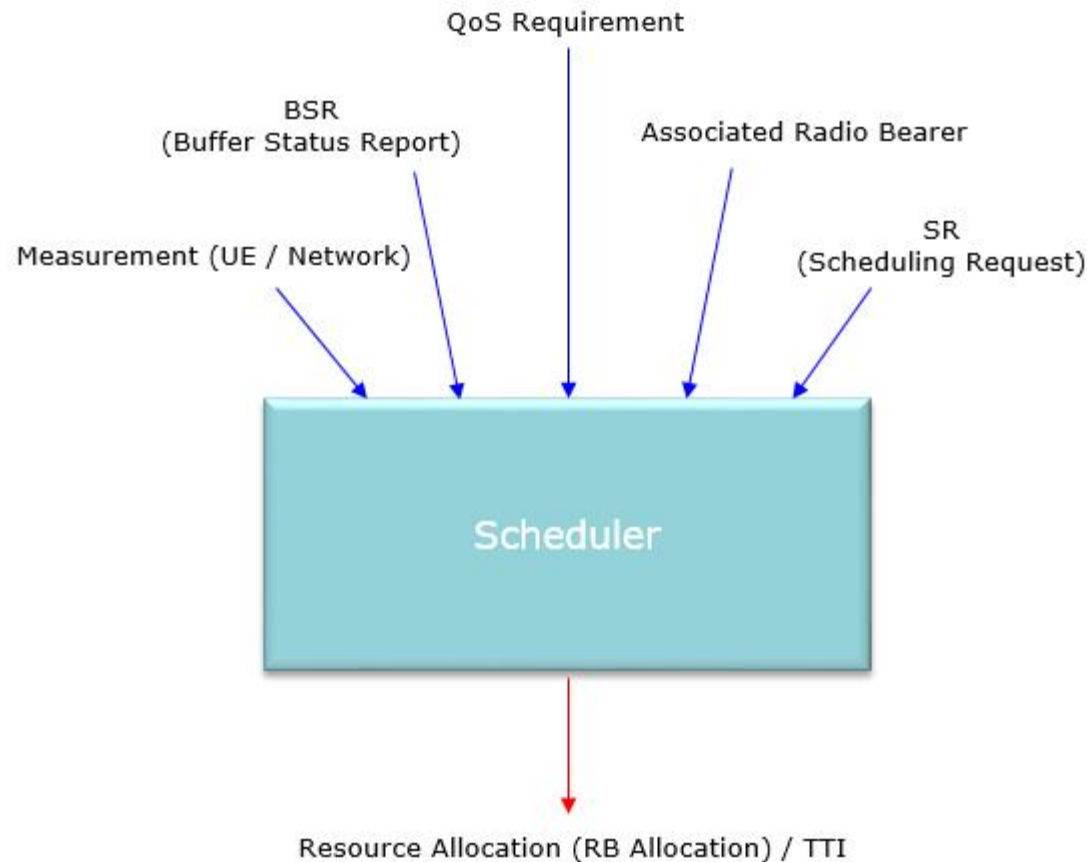
- For each QoS Flow, the QoS profile shall include the QoS parameters:
 - 5G QoS Identifier (5QI); and
 - Allocation and Retention Priority (ARP).
- For each Non-GBR QoS Flow only, the QoS profile may also include the QoS parameter:
 - Reflective QoS Attribute (RQA).
- For each GBR QoS Flow only, the QoS profile shall also include the QoS parameters:
 - Guaranteed Flow Bit Rate (GFBR) - UL and DL; and
 - Maximum Flow Bit Rate (MFBR) - UL and DL; and
- In the case of a GBR QoS Flow only, the QoS profile may also include one or more of the QoS parameters:
 - Notification control;
 - Maximum Packet Loss Rate - UL and DL.

Standardized 5QI to QoS characteristics mapping (TS23.501)

5QI Value	Resource Type	Default Priority Level	Packet Delay Budget	Packet Error Rate	Default Maximum Data Burst Volume	Default Averaging Window	Example Services
1	GBR	20	100 ms	10^{-2}	N/A	2000 ms	Conversational Voice
2		40	150 ms	10^{-3}	N/A	2000 ms	Conversational Video (Live Streaming)
3		30	50 ms	10^{-3}	N/A	2000 ms	Real Time Gaming, V2X messages. Electricity distribution – medium voltage, Process automation monitoring
79		65	50 ms	10^{-2}	N/A	N/A	V2X messages
80		68	10 ms	10^{-6}	N/A	N/A	Low Latency eMBB applications Augmented Reality
82	Delay-critical GBR	19	10 ms	10^{-4}	255 bytes	2000 ms	Discrete Automation
83		22	10 ms	10^{-4}	1354 byte	2000 ms	Discrete Automation V2X messages (UE - RSU Platooning, Advanced Driving: Cooperative Lane Change with low LoA.
84		24	30 ms	10^{-5}	1354 bytes	2000 ms	Intelligent transport systems
85		21	5 ms	10^{-5}	255 bytes	2000 ms	Electricity Distribution- high voltage V2X messages (Remote Driving).
86		18	5 ms	10^{-4}	1354 bytes	2000 ms	V2X messages (Advanced Driving: Collision Avoidance, Platooning with high LoA.
87		25	5 ms	10^{-3}	500 bytes	2000 ms	Interactive Service - Motion tracking data
88		25	10 ms	10^{-3}	1125 bytes	2000 ms	Interactive Service - Motion tracking data

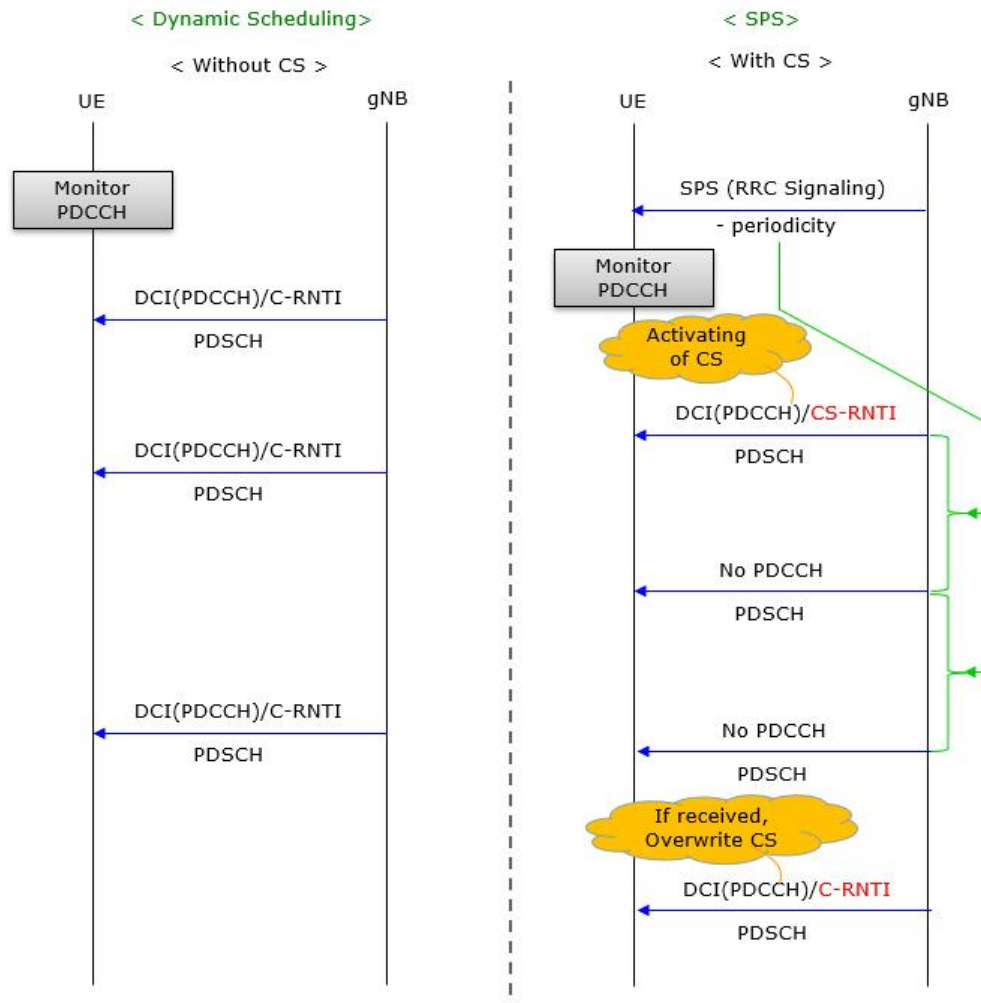
QoS Guaranteed Resource Allocation - NR Scheduling

Scheduling is the process of allocating resources for transmitting data.



NR Downlink Scheduling

There are two types of scheduling for downlink. One is called 'Dynamic Scheduling' and the other one SPS(Semi Persistent Scheduling). Dynamic scheduling is the mechanism in which each and every PDSCH is scheduled by DCI(DCI 1_0 or DCI 1_1). SPS is the mechanism in which the PDSCH transmission is scheduled by RRC message.

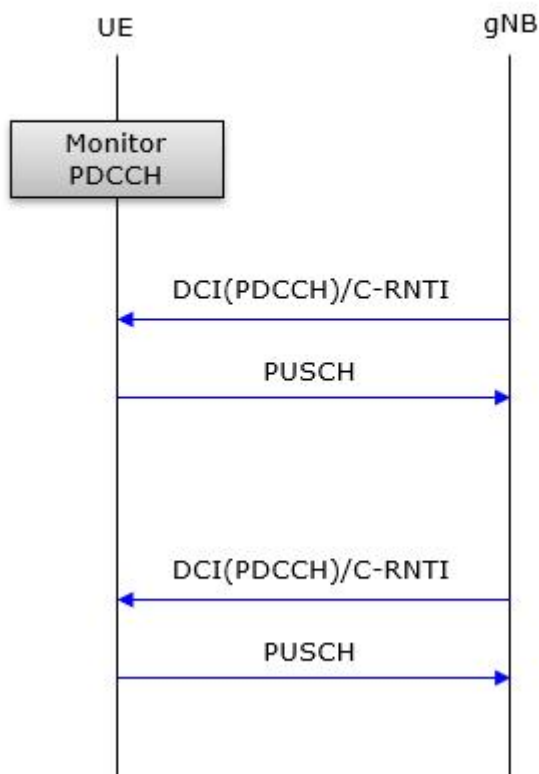


NR Uplink Scheduling

As in downlink scheduling, there are two types of scheduling for uplink. One is called '**Dynamic Scheduling**' and the other one **CS (Configured Scheduling)**'. Dynamic scheduling is the mechanism in which each and every **PUSCH** is scheduled by DCI(**DCI 1_0** or **DCI 1_1**). CS is the mechanism in which the **PUSCH** transmission is scheduled by RRC message.

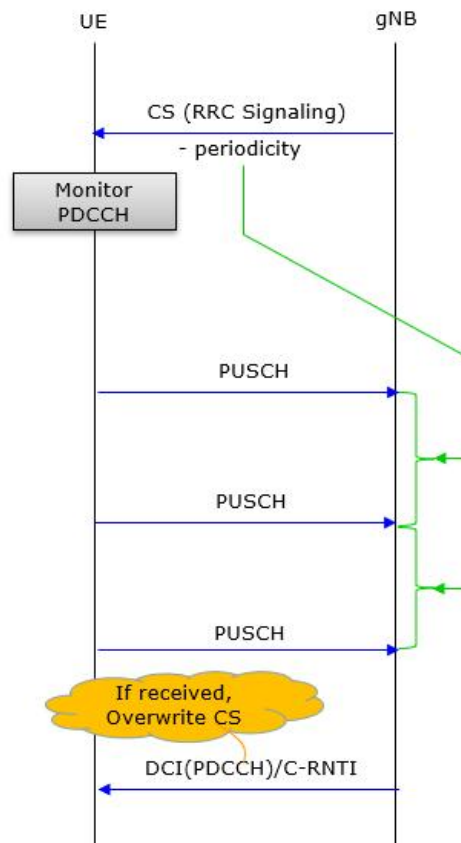
< Dynamic Scheduling >

< Without CS >

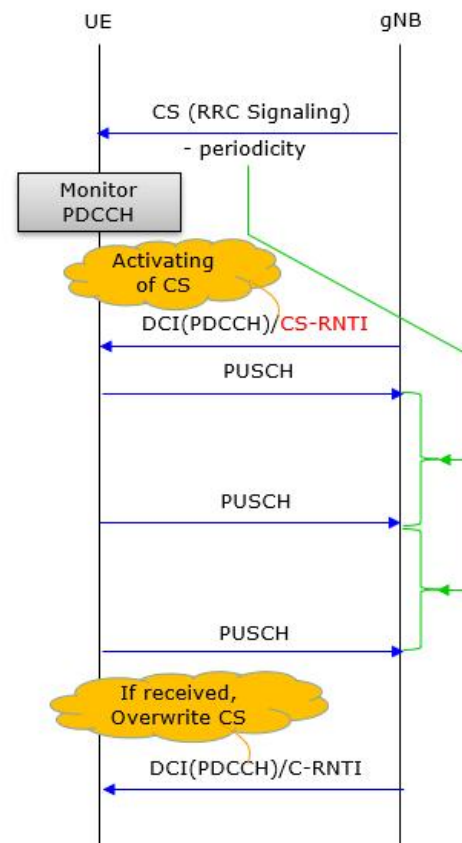


< Configured Scheduling (CS) >

< With Type 1 CS >

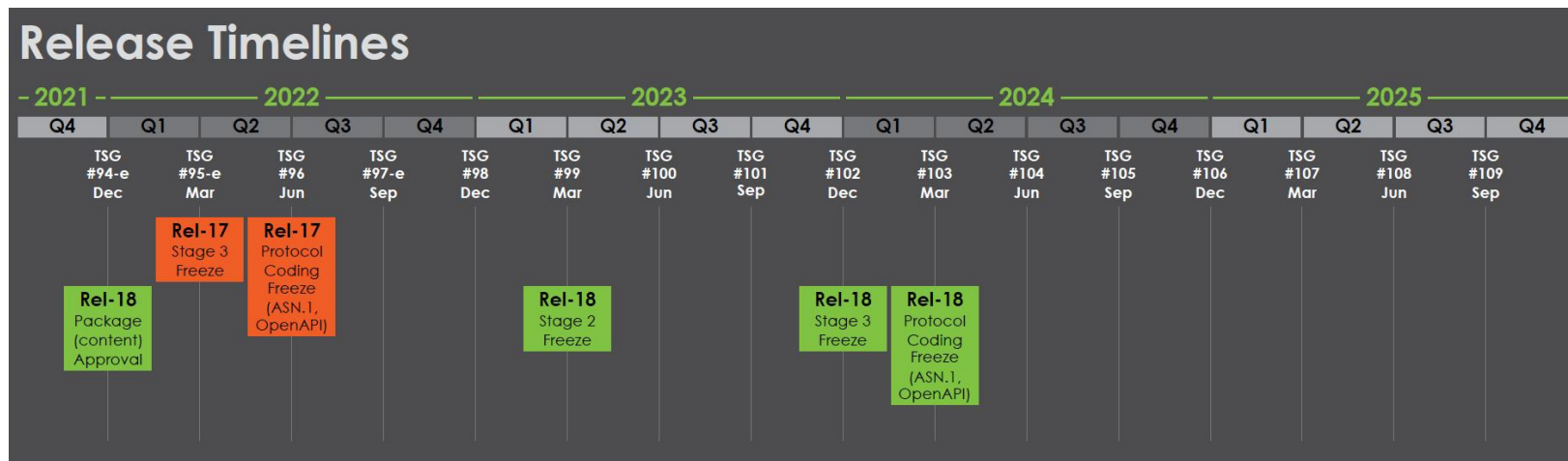


< With Type 2 CS >



3GPP Work Plan

3GPP Rel-18 Work Plan



RAN1 led - Radio Layer 1 (Physical layer)

MIMO Evolution for Downlink and Uplink
Study on Artificial Intelligence (AI)/Machine Learning (ML) for NR Air Interface
Study on Evolution of NR Duplex Operation
NR sidelink evolution
Study on expanded and improved NR positioning
Further NR RedCap UE complexity/cost reduction
Study on network energy savings
Further NR coverage enhancements
NR Network-Controlled Repeaters
Enh. of NR Dynamic spectrum sharing (DSS)
Study on low-power Wake-up Signal and Receiver for NR
Multi-carrier enhancements for NR

RAN2 led - Radio Layer 2 & Layer 3 Radio Resource Control

NR Mobility Enh.
Study on XR Enh. for NR
NR sidelink relay enh.
NR NTN (Non-Terrestrial Networks) enh.
IoT NTN enh.
NR Support for UAV
Dual Tx/Rx MUSIM
In-Device Co-existence (IDC) enh. for NR and MR-DC
Mobile Terminated-Small Data Transmission (MT-SDT) for NR
Enh. of NR Multicast and Broadcast Services

Early Release 19 Studies

SA1 - Services

Network of Service Robots with Ambient Intelligence
Energy Efficiency as service criteria
Upper layer traffic steering, switching and split over dual 3GPP access
Uncrewed Aerial Vehicles (Phase 3)
Satellite Access (Phase 3)
Roaming value added services
AI/ML Model Transfer (Phase 2)
Integrated Sensing and Communication
Ambient power-enabled Internet of Things
Localized Mobile Metaverse Services
Network Sharing Aspects
Future Railway Mobile Communication System (Phase 5)
Supporting Railway Smart Station Services

The detailed content of Rel-19 will be decided in September 2023 (TBC)

References

- Behrouz A. Forouzan, Data Communications and Networking, 机械工业出版社, 2009.
- A. Osseiran, J.F. Monserrat, P. Marsch, 5G Mobile and Wireless Communications Technology, 中国工信出版集团, 2017.
- 3GPP TS 23.501: "System Architecture for the 5G System; Stage 2".
- 3GPP TS 38.300: "NR; NR and NG-RAN Overall Description; Stage 2".
- 3GPP TS 38.331: "NR; Radio Resource Control (RRC) protocol specification".